# Nervous tissue

Prof.Dr./ Doaa Zaghloul

- The nervous tissue is composed of highly specialized cells (nerve cells or neurons) to conduct the nerve impulses and several types of supporting cells called neuroglia (glial cells).

# Neurons

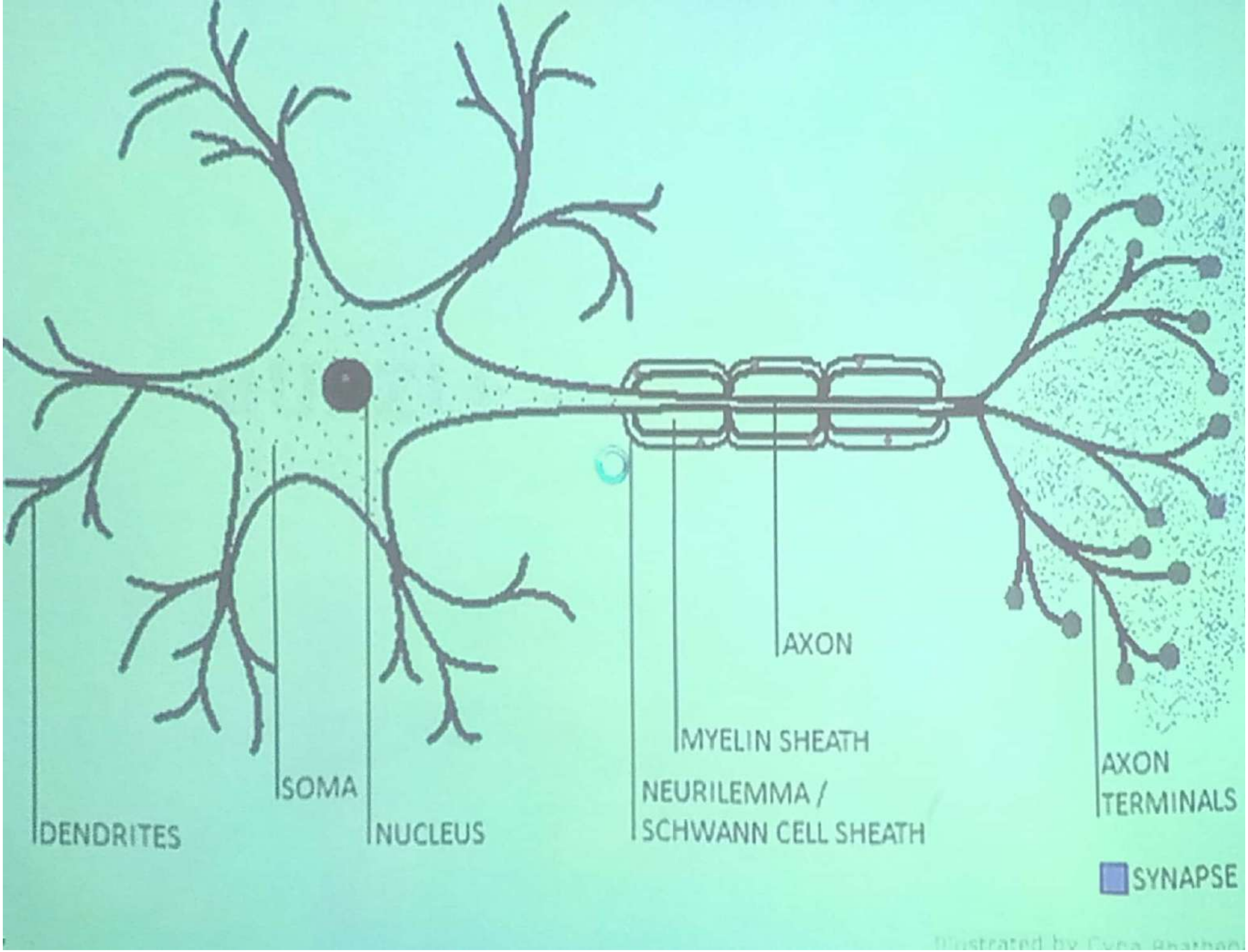
- The neuron is the structural and functional unit of the nervous tissue.
- It is responsible for the reception, transmission and processing of stimuli and the release of neurotransmitters and other informational molecules.

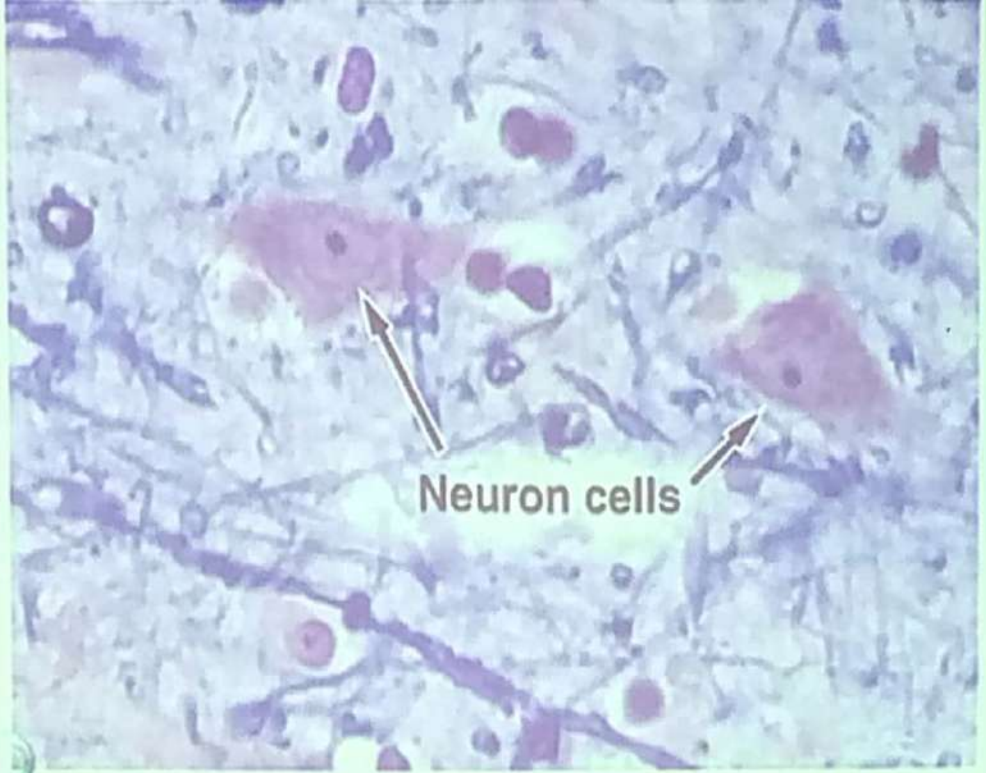
## Structure:

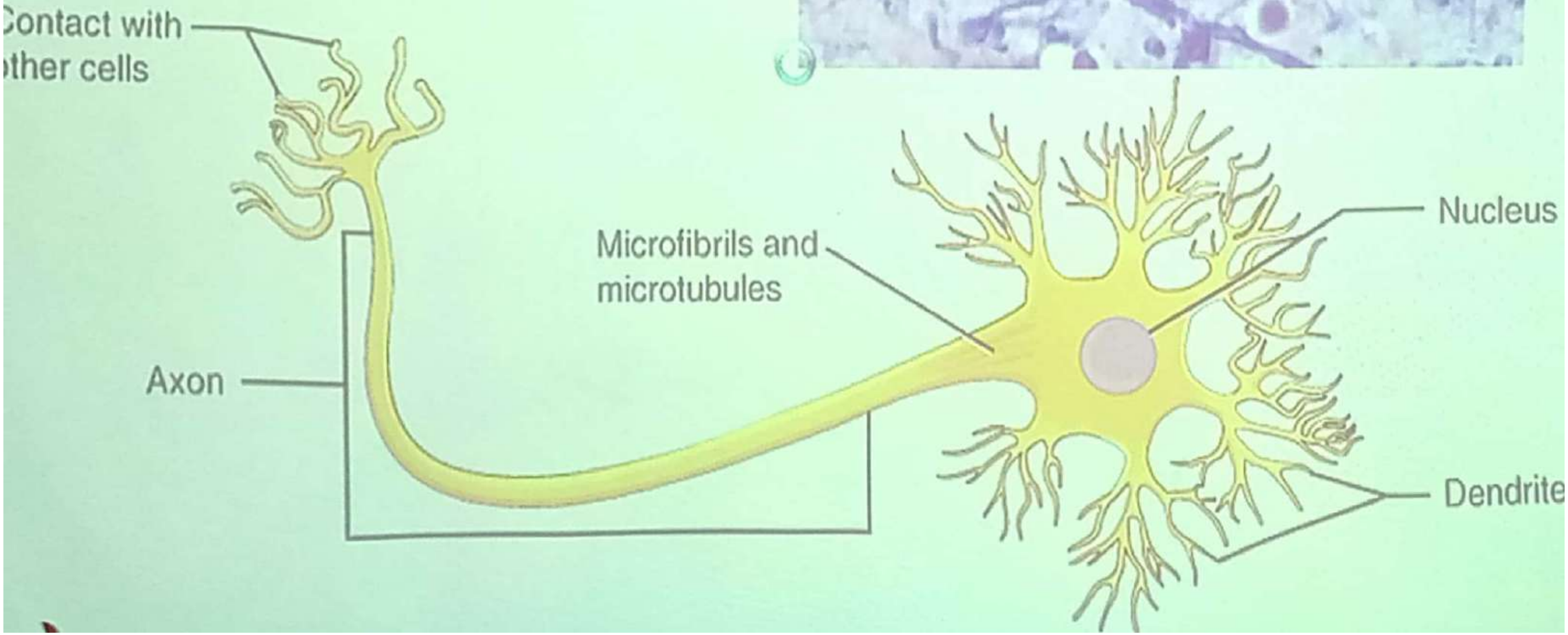
- Despit great variations in size and shape, most of the neurons have the same basic structure.

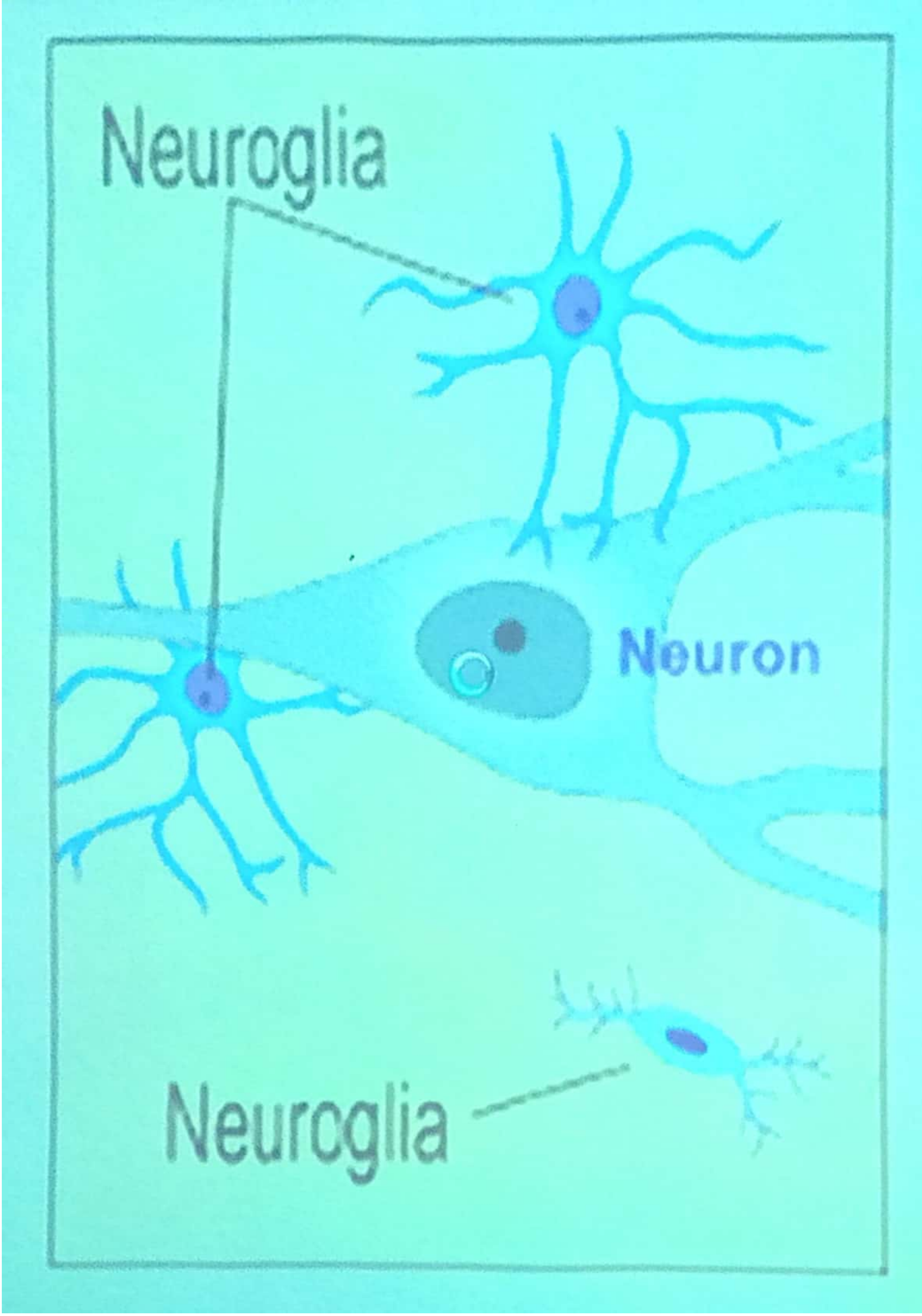
A neuron is formed of:

- Cell body
- Cytoplasmic processes extend from the cell body, namely a single axon and one or more dendrites.


AXON
MYELIN SHEATH
AXON
TERMINALS
SOMA
DENDRITES
NUCLEUS
NEURILEMMA /
SCHWANN CELL SHEATH
SYNAPSE


Neuron cells


Contact with
ther cells
Axon
Microfibrils and
microtubules
Nucleus
Dendrite


Neuroglia
Neuron
Neuroglia

## Cell body (Soma or perikaryon)

- It is the part which contains the nucleus surrounded by the cytoplasm.

- Its shape differs from neuron to another, it may be spherical, ovoid, pyramidal, stellate or pyriform.

### At LM level.

- Most neurons have a single, lightly stained spherical nucleus, centrally located with deeply basophilic prominent nucleolus.

- Binuclear nerve cells are seen in sympathetic and sensory ganglia.

- The cytoplasm has basophilic granules called Nissl bodies or granules.

- They are distributed in the cytoplasm of the cell body except in the origin of the axon (axon hillock).

- By silver stain the Golgi apparatus appears around the nucleus and neurofibrils can be seen in the cytoplasm and its processes.

## At EM level

- The nerve cell is a highly active cell and all features of this activity in both nucleus and cytoplasm are found.

- The nucleus contains chromatin mostly from the euchromatin type (extended type), so the large central nucleolus is very prominent.

- The cytoplasm contains the following organelles and inclusions.

## At EM level

- The nerve cell is a highly active cell and all features of this activity in both nucleus and cytoplasm are found.

- The nucleus contains chromatin mostly from the euchromatin type (extended type), so the large central nucleolus is very prominent.

- The cytoplasm contains the following organelles and inclusions.

1. Rough endoplasmic reticulum and free ribosomes.

- They are found in the cell body and the dendrites (not in axon and axon hillock).
- These are the basophilic <u>Nissl bodies.</u>
- They are responsible for protein synthesis (microtubules, neurotransmitters and secretion products).

2. Golgi apparatus.

- It is well developed surrounding the nucleus.
- It is the site of packing neurotransmitters into synaptic vesicles and neurosecretory vesicles.

## 3. Mitochondria.

- They are scattered all over the neuron (cell body, dendrites and axon).

## 4. Neurofilaments and microtubules.

- They are scattered all over the neuron (cell body, dendrites and axon).

- The neurofilaments have a role in maintaining the cell shape meanwhile the microtubules play a role in the active transport of proteins and organelles within the neuron.

## 5. Smooth endoplasmic reticulum.

## 6. Lysosomes and Lipofuscin pigment.

- This pigment increases with age as a residue of lysosomal activity.

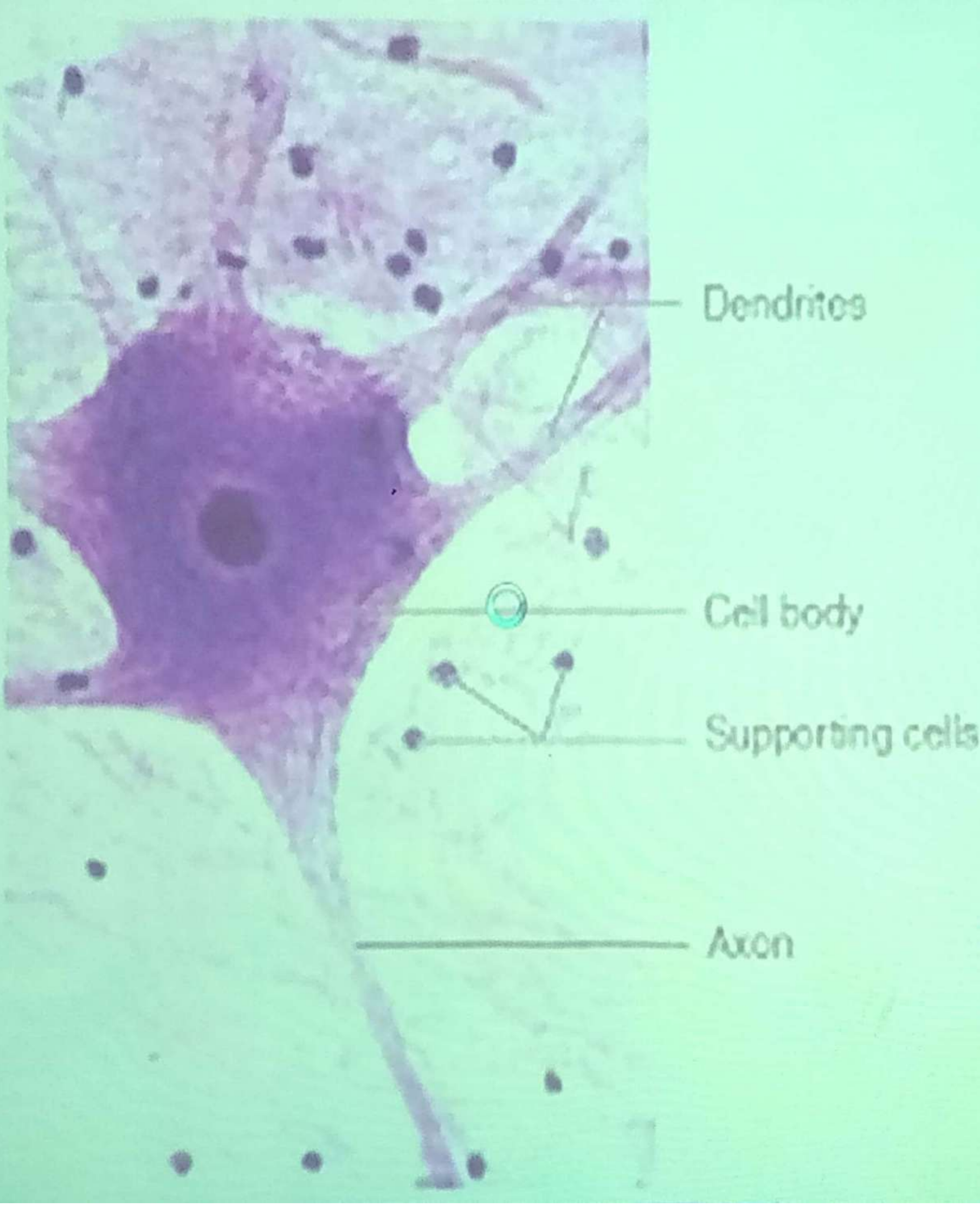
## 7. Glycogen and lipid droplets.

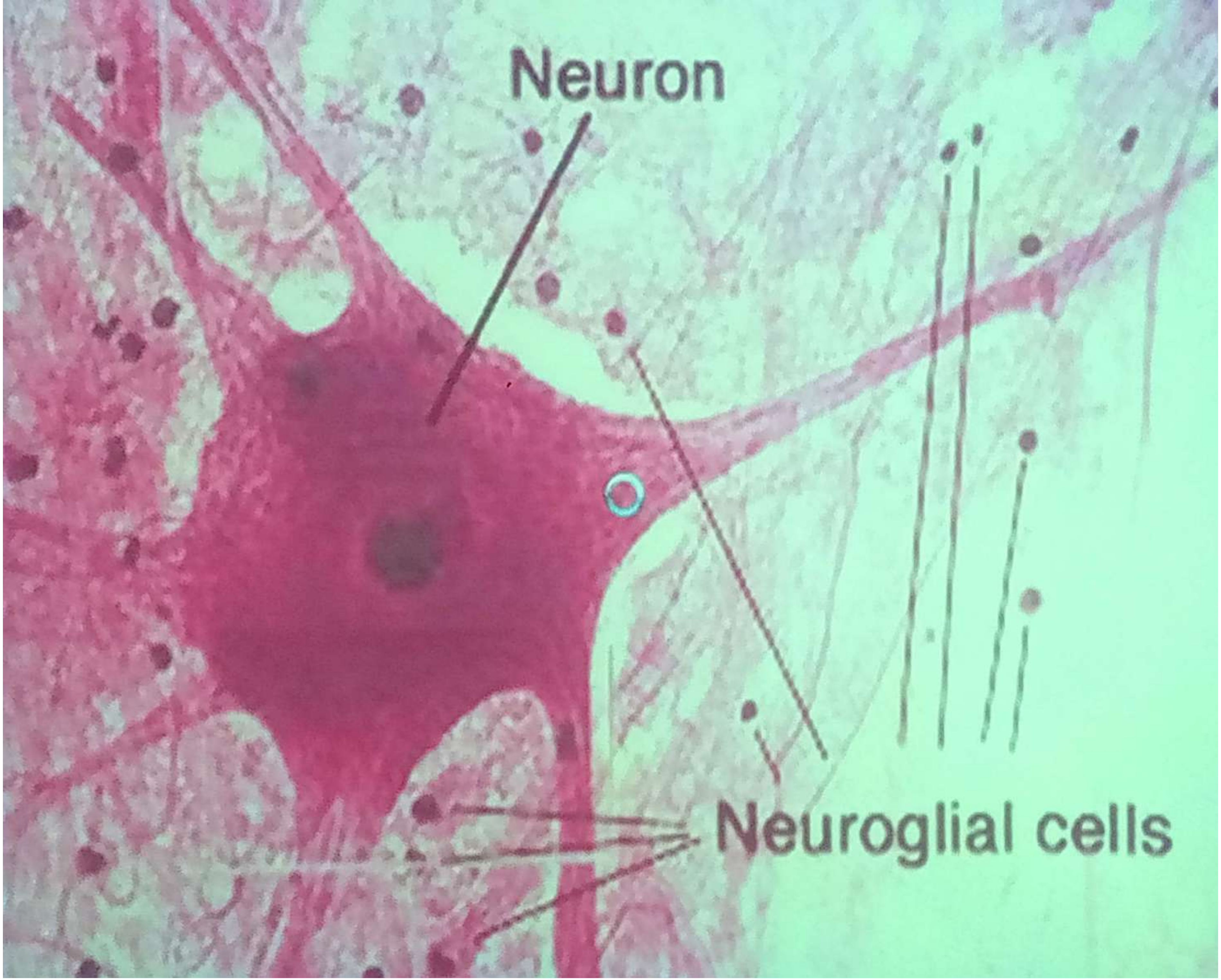
N.B. Centrosome is absent (mature nerve cell did not divide)

## Function:

The main function of the cell body is the reception of impulses from its dendrites or from other neurons and conduction these impulses via the axon to another neuron or organ.

N.B. In general, the cell bodies of all neurons are located in the gray matter of the CNS except the cell bodies of primary sensory neurons (olfactory and taste bud neurons) and gangalionic neurons.


Dendrites
Cell body
Supporting cells
Axon


Neuron
Neuroglial cells

- They are tree-like extensions of the cell body to increase the surface area for impulse reception.

- They contain most of the cell organelles and inclusions as the cell body but lack Golgi apparatus.

- Their surfaces are covered by small projections called dendritic spins or gemmules which act as synaptic sites.

- They receive the impulses and carry them toward the cell body.

# Dendrites

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- They contain most of the cell organelles and inclusions as the cell body but lack Golgi apparatus.

- Their surfaces are covered by small projections called dendritic spins or gemmules which act as synaptic sites.

- They receive the impulses and carry them toward the cell body.

# Axons

- It is a single cylindrical process, varies in length and diameter according the type of neuron.
- All axons originate from short pyramidal-shaped region, the axon hillock.
- It branch at its end into terminal arborization.
- The axonal cytoplasm (axoplasm) contains mitochondria, microtubules, neurofilaments and some smooth ER but no Nissl bodies.

- Few side branches called collateral may come out at right angels from the axon.

- It conduct impulses away from the cell body to other neurons or organs and also generate impulses.

- It is covered by myelin sheath in myelinated nerve fibers and by neurolemma in most of them.

## Classification of neurons

A- Accoding to number of cell processes, neurons are classified into:

### 1. Unipolar neurons

- Have one process (axon) and no dendrites.
- Site: Photoreceptor cells (rods and cons)

### 2. Pseudounipolar neurons

- - Have single process close to the cell body divides into two branches forming T- shape. One branch becomes a dendrite and the other becomes an axon.
- Site: Spinal ganglia.

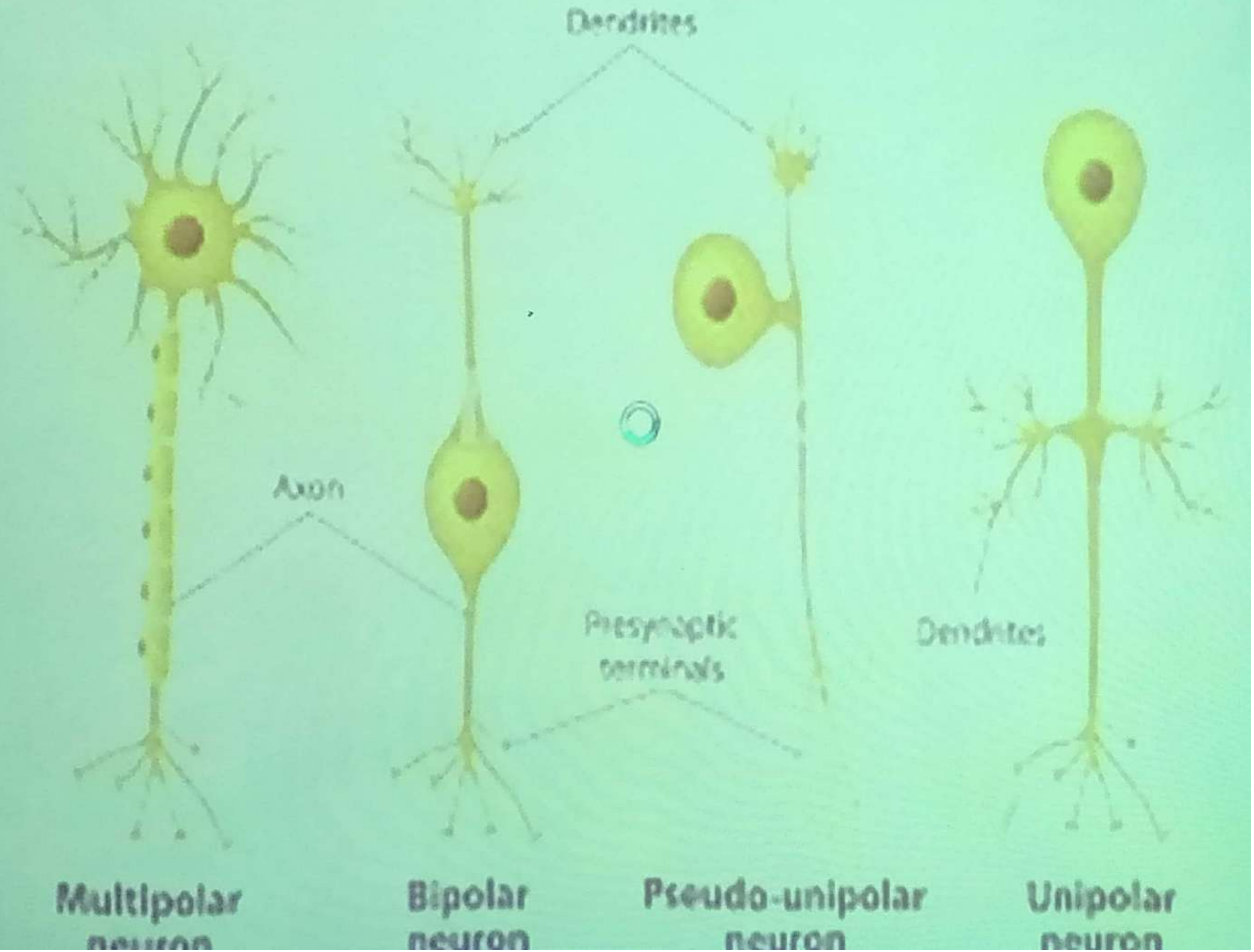
### 3. Bipolar neurons

- Have two processes one axon and one dendrite.
- Site: Olfactory mucosa and retina.

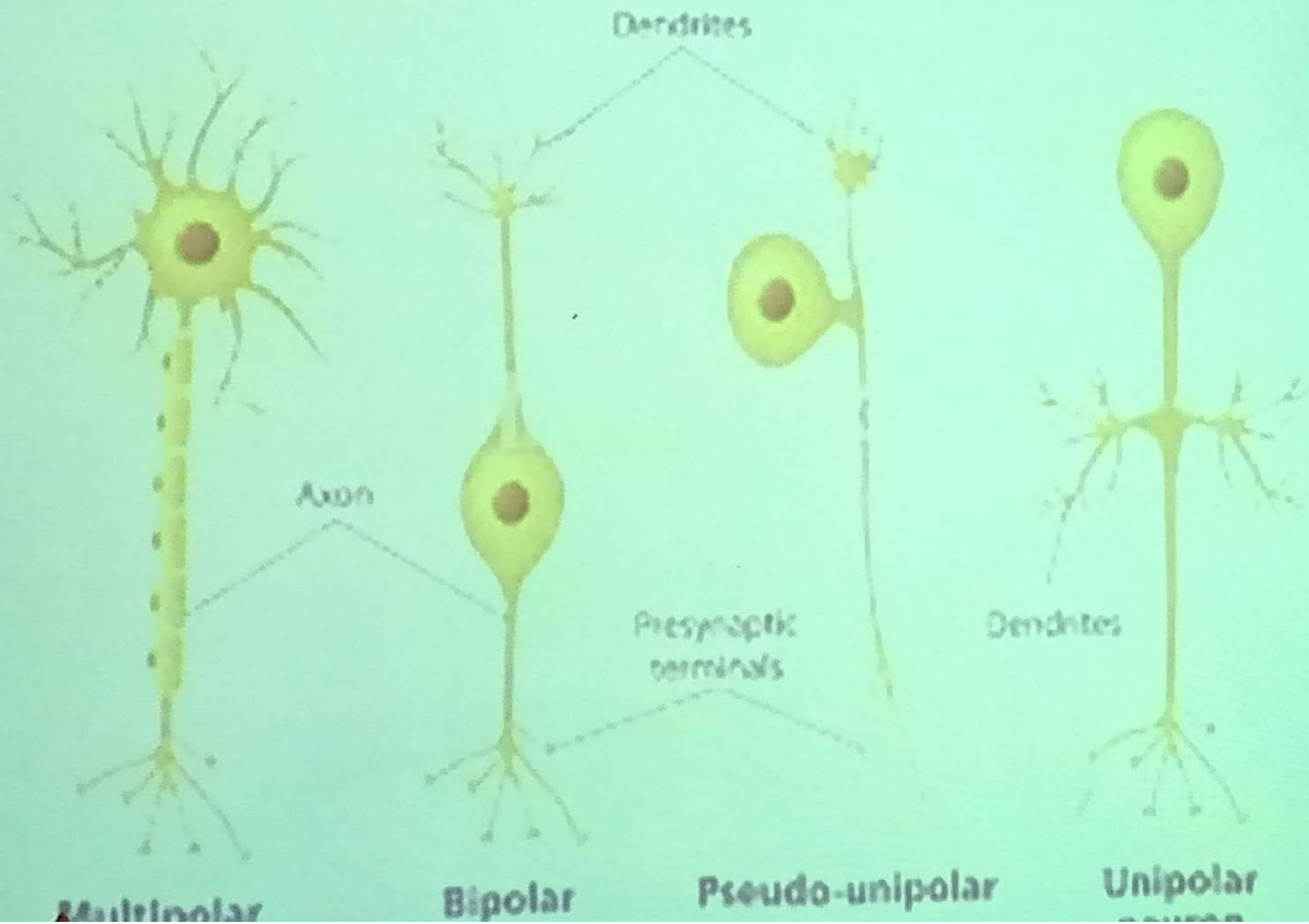
### 4. Multipolar neurons

- Have several processes arise from the cell body, several dendrites and one axon.
- Site : Most of neurons eg. Motor neurons of spinal cord

# Basic Neuron Types



# Basic Neuron Types



B. **According to the size and length of axon**, neurons are classified into:

**1. Golgi type I neurons**

- Have a large cell body and long axon.
- Site: Motor neurons of spinal cord.

**2. Golgi type II neurons**

- Have a small soma and short axon.
- Site: Interneurons of spinal cord.

C- According to the function, neurons are classified into:

1. Motor neurons

- Carry motor impulses from CNS to peripheral end organs.
- Site : Pyramidal cells of cerebral cortex.

2. Sensory neurons

- They receive impulses from peripheral sensory cells and organs and carry them toward CNS.
- Site: Sensory spinal ganglia.

3. Interneurons

- They are small and short neurons connect the sensory and motor neurons.
- Site: Golgi type II in the spinal cord and brain.

# Synapse

- Synapse is specialized junction by which a stimulus (impulse) is transmitted from a neuron to another or to effector cell (muscle and gland cells).

**1- According to the method of impulse transfer, synapses classified into:**

1- Chemical synapses
2- Electric synapses

**2- According to the site of synapse, synapses are classified into:**

1- Axo-dendritic: Between axon and dendrite.
2- Axo-somatic : Between axon and cell body.
3- Axo-axonic : Between axon and axon

# Chemical synapses

- It is the most common type.
- Impulse is transmitted by releasing chemical substances (neurotransmitters) at the axon terminal to induce the transfer of the impulse to other neuron or to effector cells.
- More than 35 neurotransmitters are known eg. acetylcholine, adrenaline, noradrenaline, dopamine, seratonin, ......

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## ❖ Structure of synapse

3 major structural components of each synapse:

1. Presynaptic membrane:

- ➢ It is a thickened electron-dense membrane of the terminal button closest to the target neuron.

- ➢ On stimulation, synaptic vesicles in the button fuse with the membrane and exocytose their neurotransmitters into the synaptic cleft.

- After the release of the neurotransmitters into the cleft, they bind with the receptors causing depolarization to the postsynaptic membrane (transfer of the impulse).

- This synapse is called excitatory synapse.

- On the other hand inhibitory synapse cause hyperpolirization to the postsynaptic membrane with no transmission of nerve impulse

- Types of chemical synapse: According type of impulse membrane

1- Excitatory synapse:

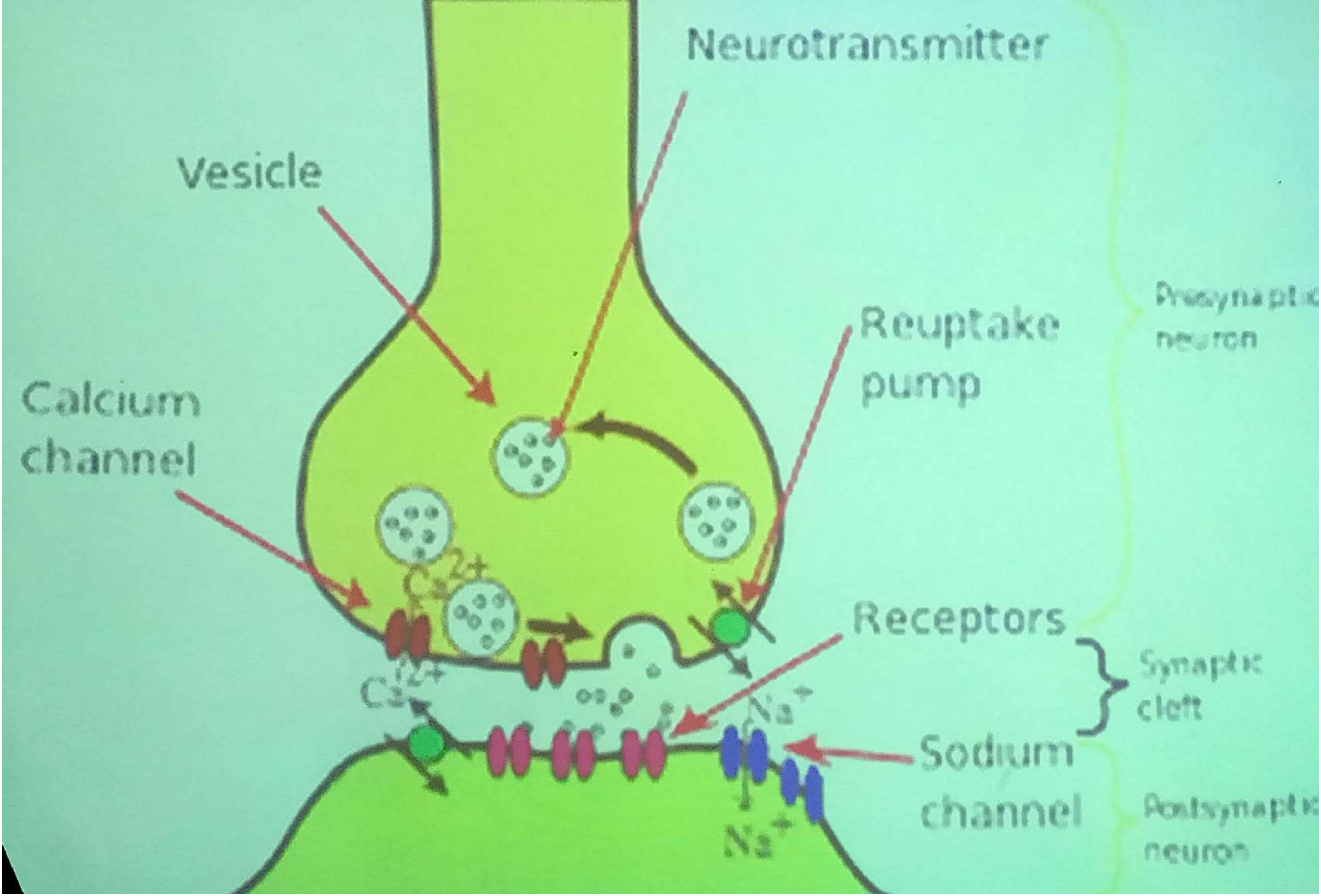
It cause depolarization to the postsynaptic.

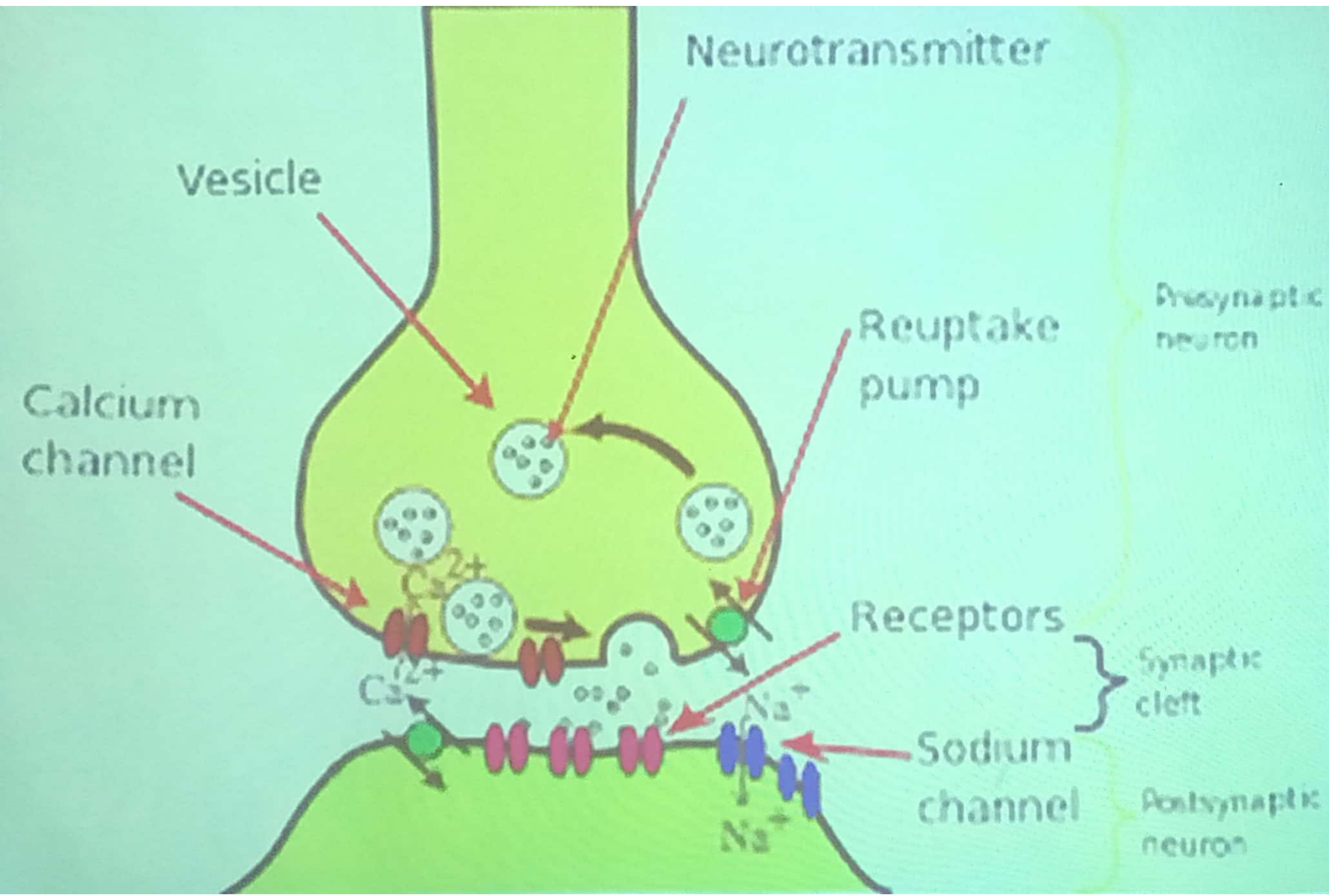
2- Inhibitory synapse:

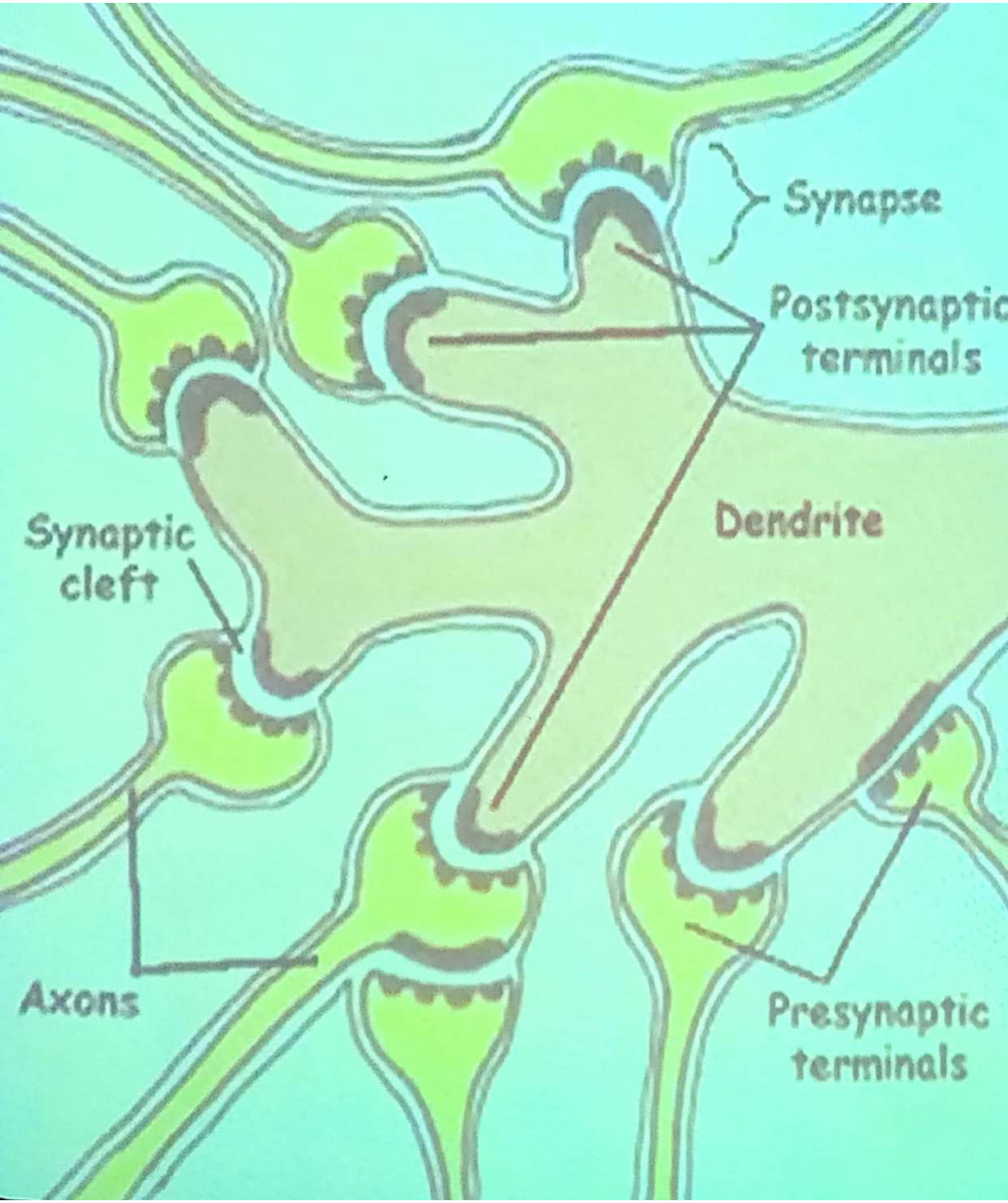
It cause hyperpolirization to the postsynaptic with no transmission of nerve impulse.

## Electric synapses

- It is very few and transmit the impulse through gap junction that cross the pre-and postsynaptic membranes.
- Ions pass freely through the gap junction and conduct the nerve impulses directly.


Neurotransmitter
Vesicle
Reuptake pump
Calcium channel
Receptors
Sodium channel
Presynaptic neuron
Synaptic cleft
Postsynaptic neuron
Na+
Na+


Neurotransmitter
Vesicle
Reuptake pump
Presynaptic neuron
Calcium channel
$Ca^{2+}$
Receptors
Synaptic cleft
$Na^+$
Sodium channel
Postsynaptic neuron


Synapse
Postsynaptic terminals
Dendrite
Synaptic cleft
Axons
Presynaptic terminals

## Supporting cells (Neuroglia)

- These cells play an important role in supporting the neurons.

- They are 10 times more abundant than neurons, surround their cell bodies and processes and occuping the interneural spaces.

- They are smaller than neurons and their processes need special stain, so identification of these cells usually based on their nuclear morphology.

- They are found in both central and peripheral nervous systems.

## Neuroglia of central nervous system (CNS)

### 1- Astrocytes

- They are of ectodermal origin
- There are two types of astrocytes:

A. Protoplasmic Astrocyte

- Site: Gray matter
- They are large branched cells with large ovoid-shaped lightly stained nuclei centrally located.

- The cytoplasm is granular (gliosomes) representing mitochondria.

- The cell processes are numerous, thick, wavy and branched.

- One or more of these processes end on the surface of blood capillary forming perivascular feet.

## B. Fibrous Astrocytes

- Site : White matter

- They resemble the protoplasmic, but their processes are thinner, straighter, longer, less branched and contain more microfilaments.

- They end in perivascular feet.

# Functions of Astrocytes

1- They control the exchange of materials between the neuron and blood streem forming a Blood Brain Barrier.

2- They control the ionic and chemical enviroment of the neurons.

3- They release metabolic substances and neuroactive molecules which influence neural survival and activity.

4- When CNS is damaged, astrocytes proliferate to form scar tissue.

# Blood-brain barrier

- It is a functional barrier that prevent the passage of some substances such as antibiotics and chemical and bacterial toxic matter from the blood to nervous tissue.
- It results by reducing the permeability from the blood capillaries.

- It is formed from:

1- Occluding junctions between the endothelial cells of the blood capillaries.
2- Non-fenestrated blood capillaries.
3- Expansions of neuroglial cell processes (perivascular feet) that envelop the blood capillaries.

## 2. Microglia (mesoglia)

- They are of mesodermal origin.
- Site :Gray and white matters.
- They are small, spindle shaped cells with elongated deeply stained nucleus.
- A thin highly branched cytoplasmic process arise from each pole of the cell.

**Functions:**

1- They are a phagocytic cells representing the mononuclear phagocytic system in nervous tissue.

2- They are involved with inflammation and repair of CNS.

3- They produce neutral proteases and oxidative radicals.

## 3. Oligodendroglia

- They are of ectodermal origin.
- Site: White and gray matters.
- They have small cell body with few short and less branched processes

**Functions:**

1- They from the myelin sheath in CNS (white matter).

2- A single cell could myelinate more than one axon.

3- In gray matter they serve as perinural satellite cells with unknown function

## 4. Ependymal cells:

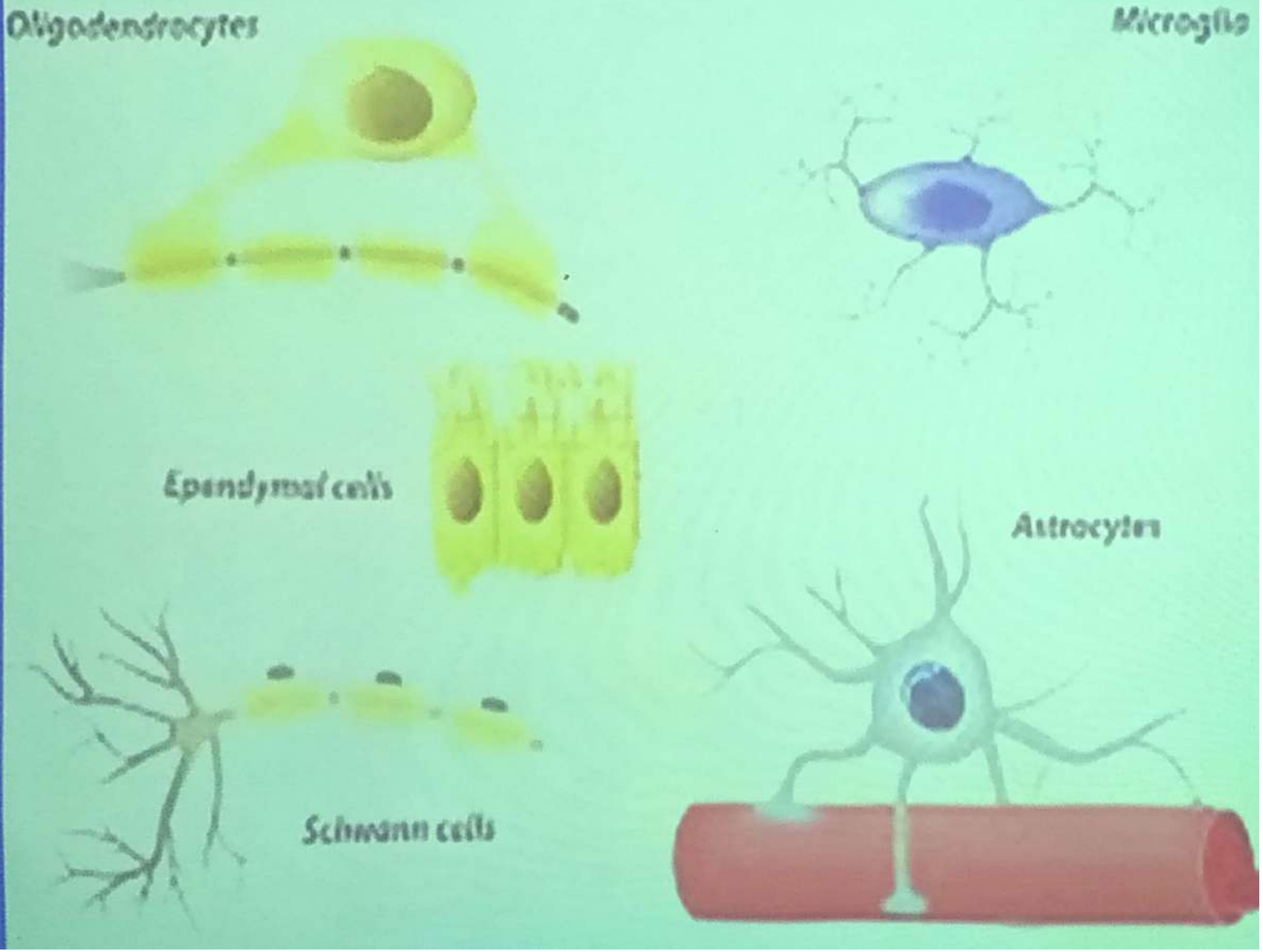
- They are of ectodermal origin.
- They are low columnar ciliated epithelial cells.
- Site: They line the cavities of CNS (ventricle of brain and central canal of spinal cord).

**Functions:**

1- They act as diffusion barrier between the cerebrospinal fluid and the neurons of CNS.

2- They control the amount of cerebrospinal fluid by absorbing or secreting fluid.

# GLIAL CELLS



Oligodendrocytes
Microglia
Ependymal cells
Astrocytes
Schwann cells

## Neuroglia of peripheral nervous system (PNS)

### 1- Schwann cells (lemmocyte)

- The same function as oligodendroglia but one Shwann cell forms myelin sheath around a segment of one axon, in contrast oligodendroglia branch to serve more than one axon.
- The myelin sheath is formed from many layers of Shwann cell plasma membrane and the cytoplasm is squeezed to forms the outer schwann sheath (neurolemmal sheath).

### 2- Satellite cells

- They are specialized cells found in the ganglia in which they form a thin layer (capsule) covering the nerve cell bodies.

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# Nervous System

Anatomically: the nervous system is divided into:

1. Central nervous system, consisting of:
   - A- Brain
   - B- Spinal cord
2. Peripheral nervous system, consisting of:
   - A- Nerve fibers (Nerves)
   - B- Ganglia
   - C- Nerve endings

## Functionally:

the nervous system can be divide into:

1- Central nervous system
2- Peripheral nervous system
3- Autonomic nervous system, which divide into:

A- Sympathetic nervous system
B- Parasympathetic nervous system

## Central Nervous System (CNS)

- The central nervous system consists of Brain (cerebrum and cerebullum) and spinal cord.

- When CNS is sliced, one can identify white matter, gray matter and regions of mixed gray and white matters.

* **Gray matter is formed from:**

1. Nerve cell bodies (nuclei)
2. Dendrites and the proximal naked unmyelinated axons.
3. Neuroglia (protoplasmic astrocytes and microglia).

* **White matter is formed from:**

1. Dense accumulation of myelinated axons.
2. Neuroglia (fibrous astrocytes, oligodendroglia and microglia).

**N.B.**

- The high lipid content of myelin is responsiple for the white appearance.
- No nerve cell bodies in the white matter.
- The gray matter on the surface of the cerebrum and cerebullum is called cortex

Both the brain and the spinal cord are enclosed and protected by fibrous membranes, the meninges.

## Meninges

- It is composed of dura mater, arachnoid and pia mater.

## Dura mater

- It is the external layer, composed of dense C.T. continuous with the periosteum of the skull.

- In the spinal cord it is separated from the periosteum of the vertebrae by the epidural space which contains loose C.T. and adipose tissue

- It is separated from the arachnoid by thin subdural space.
- It is internal surface is covered by simple squamous epithelium.

## Arachnoid

-It is composed of C.T. devoid of blood vessels.

-It has two components, a layer in contact with the dura mater and a system of trabeculae connecting this layer with the pia mater.

-The cavities between the trabeculae form the subarachnoid space which is filled with cerebrospinal fluid and communicates with the ventricles of the brain.

-The surfaces of the arachnoid is covered by simple squamous epithelium.

-In some areas the arachnoid perforates the dura mater forming protrusions (arachnoid villi) that terminate in venous sinuses. These protrusions are covered by endothelial cells of the veins.

-The function of the villi is to resorb cerebrospinal fluid into the blood venous sinuses.

## Pia mater

-It is a loose C.T. containing many blood vessels.

-It is separated from the nerve cells and fibers by a thin layer of neuroglial processes which form a physical barrier.

-It is covered by simple squamous cells.

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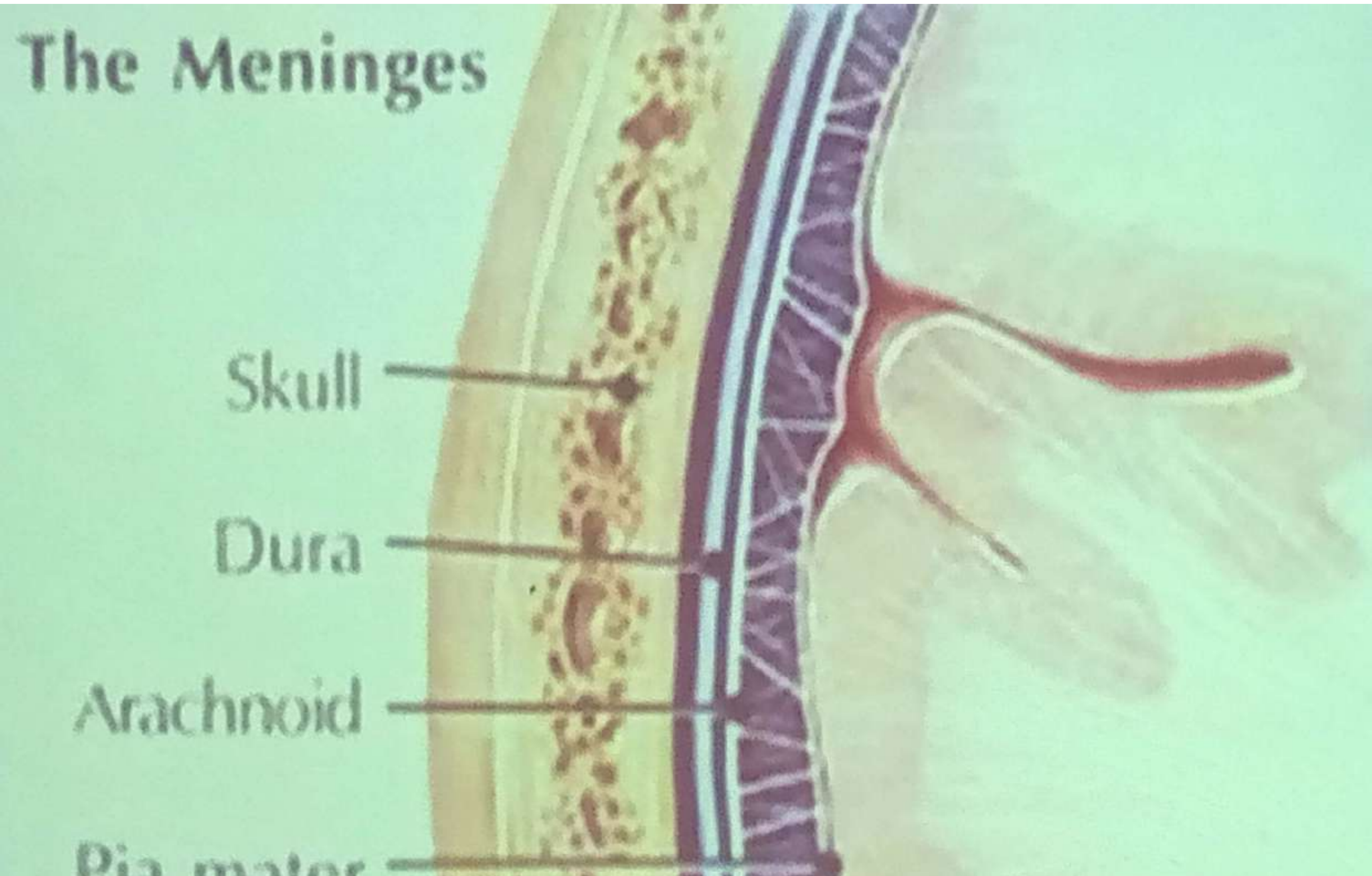



Fig. 2

## Cerebrum

- ✓ It is composed of paired cerebral hemispheres.
- ✓ The surface of each hemispher has gyri (ridges) demarcated by sulci (grooves).
- ✓ The gray matter (cerebral cortex) can be divided into six layers. From superficial to deep, these layers are:

1- Molecular layer : contains few cell bodies and many fibers.

2- External granular layer : contains many small nerve cells.

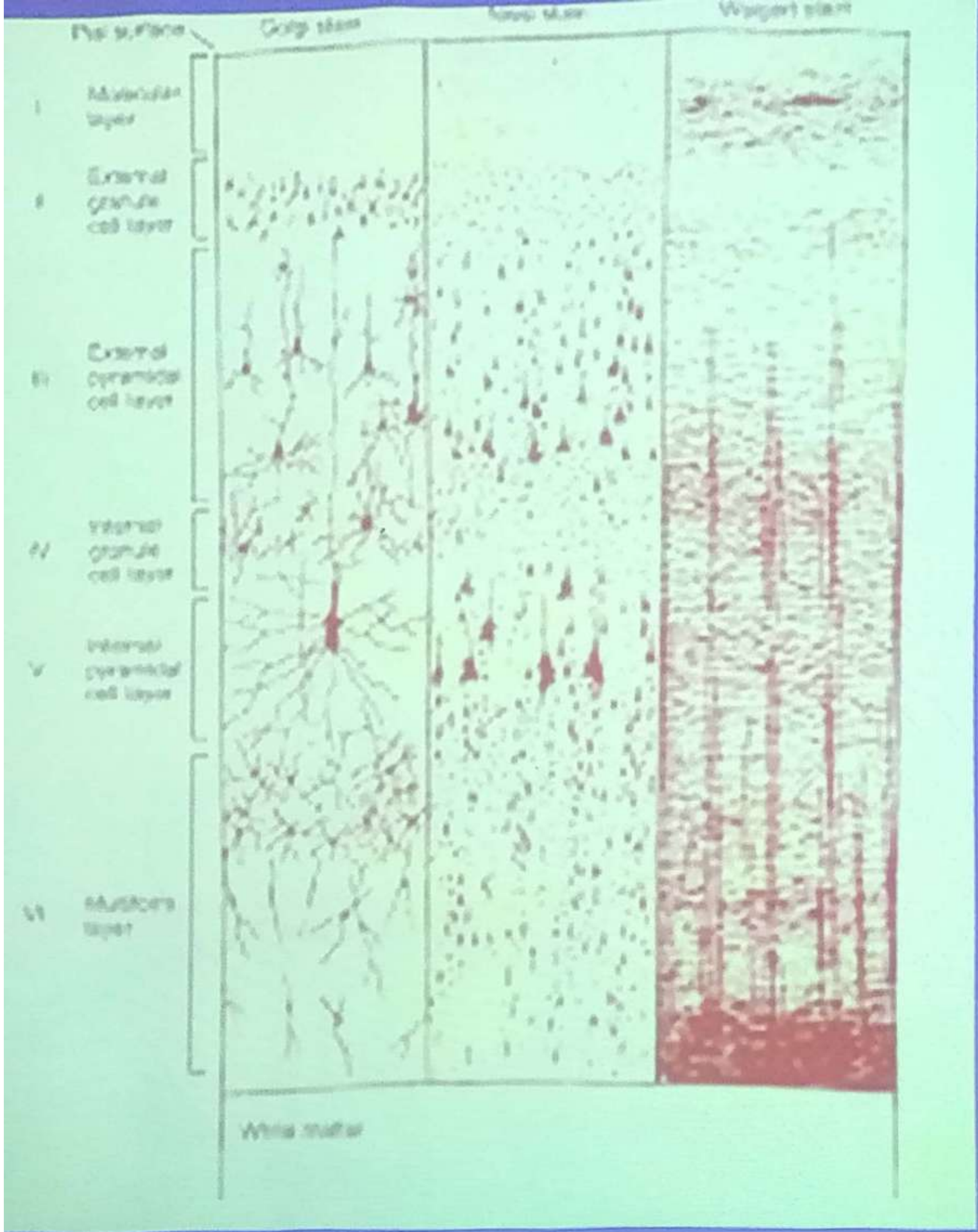
3- External pyramidal layer : contains medium and large pyramidal neurons.

4- Internal granular layer : contains many small neurons (stellate). It is the primary receptive layer (sensory) eg. visual area.

5- Internal pyramidal layer : contains medium to very large pyramidal neurons. It is the cortical motor area.

6- Fusiform (multiform) layer : contains many spindle-shaped neurons as also nerve cells with many shapes.

- ✓ Deep to the last layer is the cerebral white matter which is composed of nerve fibers going to and coming from the cortex.

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# Cerebellum

- ✓ It consists of two cerebellar hemispheres connected with a narrow median vermis
- ✓ It is surface has narrow ridges (folia) separated by sulci (grooves).
- ✓ The gray matter (cerebellar cortex) can be divided into 3 layers:

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- ✓ It is surface has narrow ridges (folia) separated by sulci (grooves).
- ✓ The gray matter (cerebellar cortex) can be divided into 3 layers:

1-Molecular layer: outer layer and consists of:-

a- Stellate cells.

b- Basket cells.

c- Neuroglia cells.

d- Dendrites of pyrkinji cells.

e- Axons of granular cells.

f- The ends of climbing fibers.

2-Purkinji cell layer:

- Pear-shaped cells arranged in one layer.
- Their dendrites form a tree-like that form the molecular layer and their axons pass through granular layer to enter white matter and end around the cerebellar nuclei

## 3-Granular layer:

- Many small dark nerve cells.
- The dendrites make synapse with the afferent Mossy fibers (from brain stem and spinal cord).
- Axons run to the molecular layer.
- Golgi type II cells are scattered in between the granular cells.
- Climbing afferent fibers originates from inferior olive pass through the granular layer to terminate on Purkinji dendrites.
- The white matter (medulla) is formed from:

i. Efferent fibers (axons of Purkinji cells).

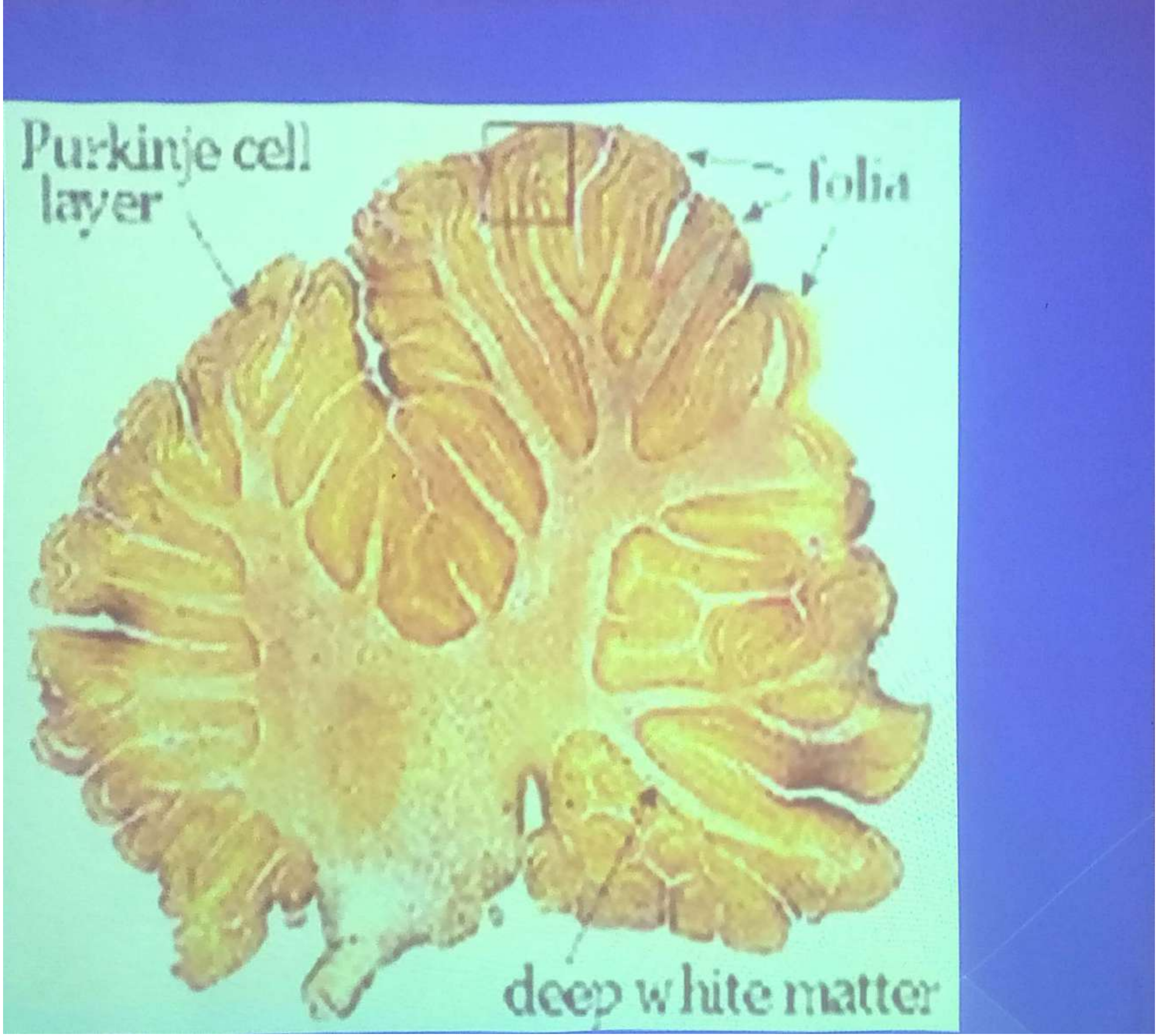
ii. Afferent fibers: - Climbing fibers

- Mossy fibers.

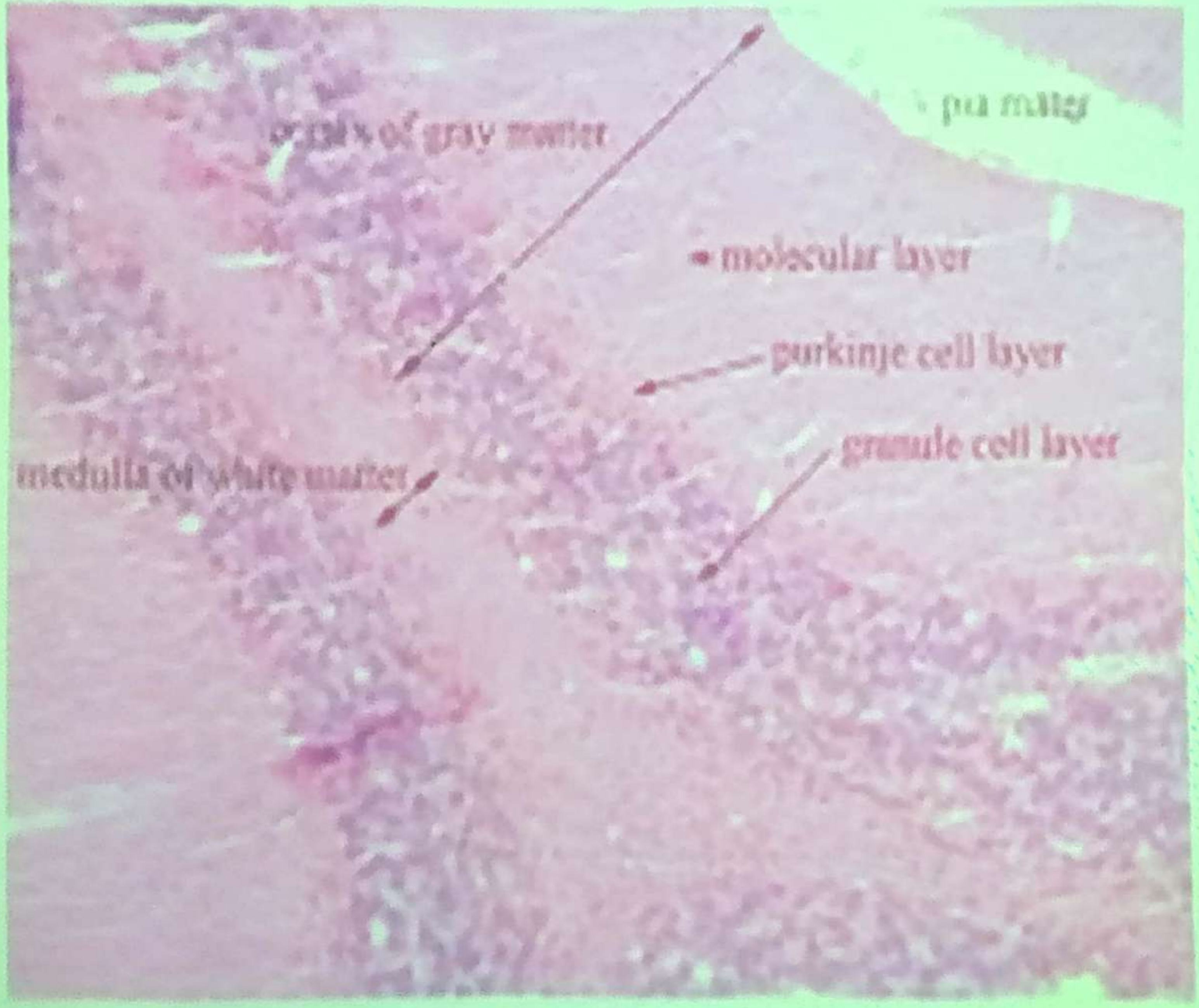
## Function

1-It regulates the movements of the body.

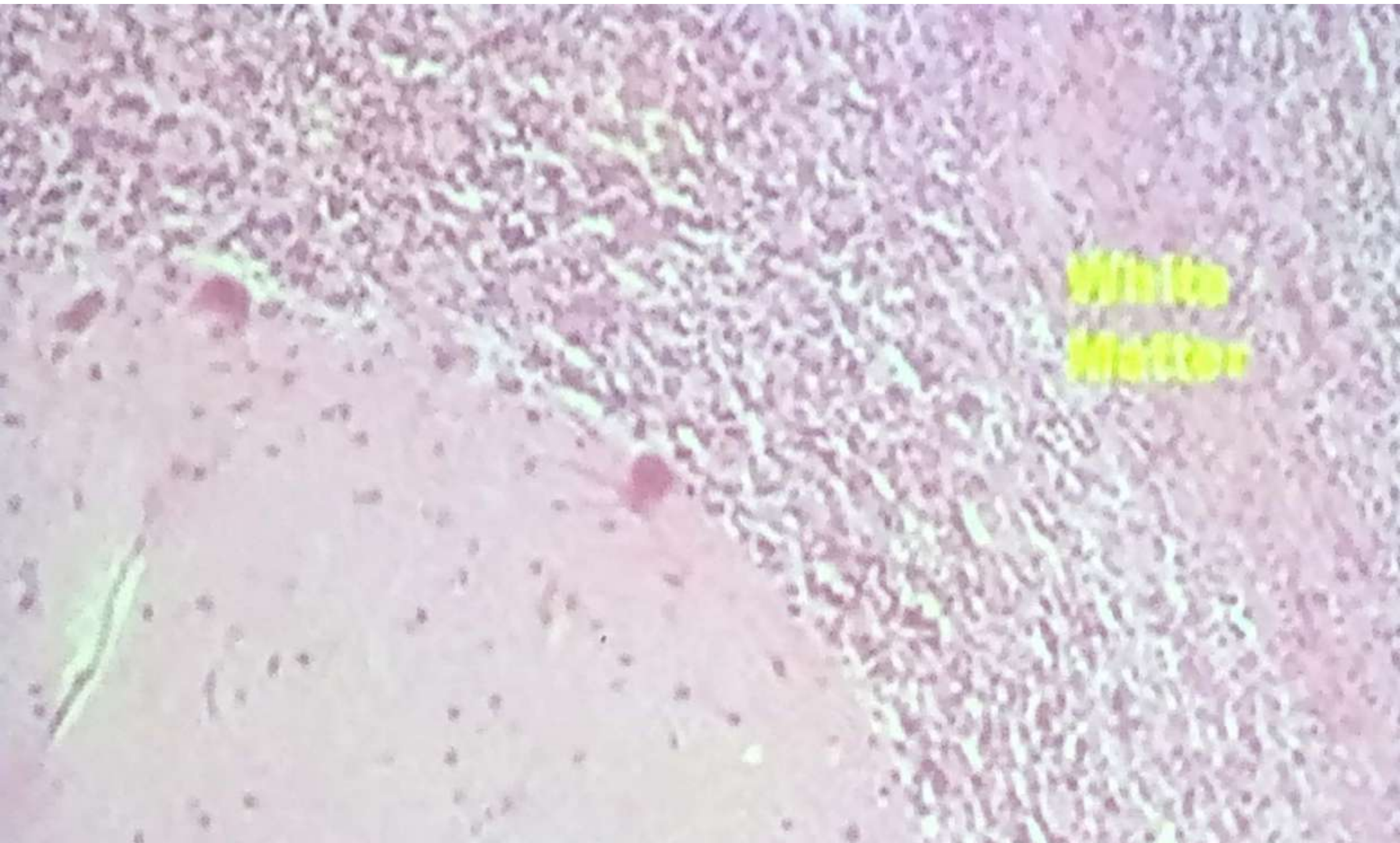
2-It coordinates the contractile activity in the different groups of skeletal muscle making the movement smooth.

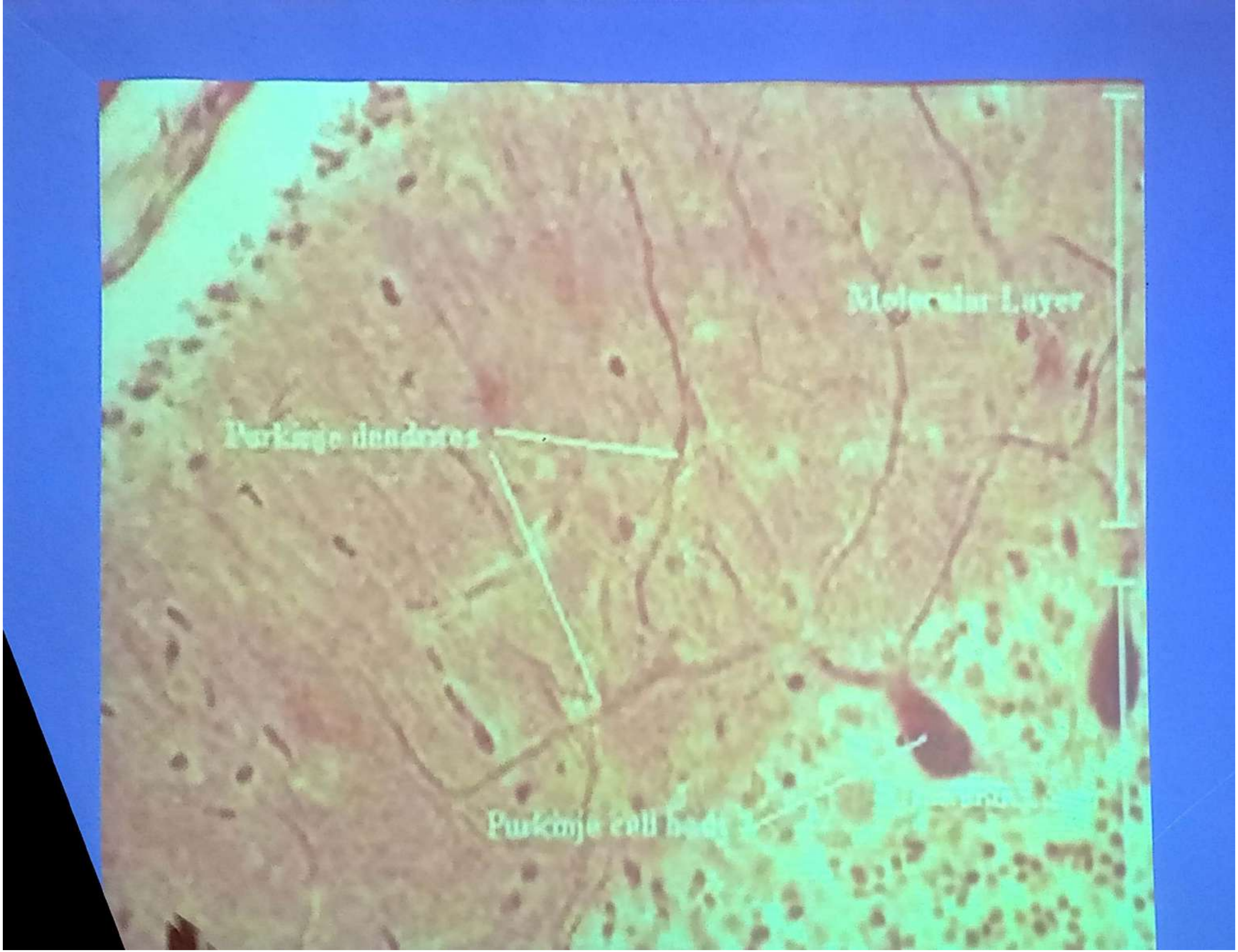

Purkinje cell layer
folia
deep white matter

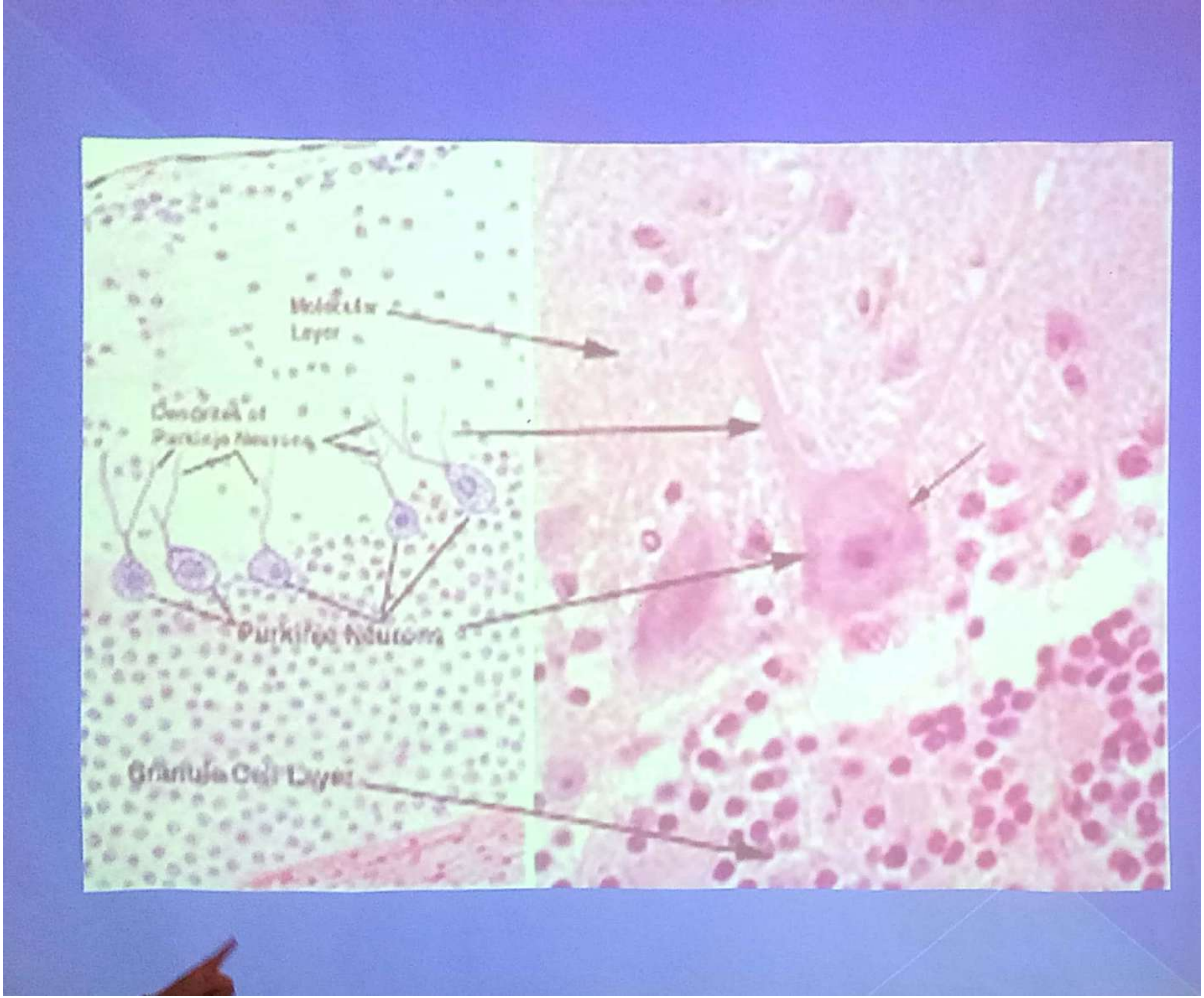
# HISTOLOGICAL STRUCTURE OF CE




human cerebellum


White Matter
Molecular Layer
Purkinje Layer
Granular Layer


Molecular Layer
Purkinje dendrites
Purkinje cell body


Layer
Granule Cell Layer

## Spinal cord

- ✓ It is that portion of the CNS caudal to the brain and is contained in the vertebral canal.
- ✓ Transverse section of the spinal cord reveals a central canal surrounded by a H-shaped area of gray matter, which in turn is surrounded by white matter.
- ✓ The spinal cord is divided dorsally by dorsal median septum and ventrally by ventral median fissure.

- ✓ The ventral gray horn contain somatic efferent large neurons that innervate the skeletal muscles (motor neurons).

- ✓ The dorsal gray horn contains small neuron and interneuron (sensory neurons).

- ✓ Lateral horn of the thoracolumber segments have medium size neuron (visceral efferent neurons that synapse in autonomic ganglia).

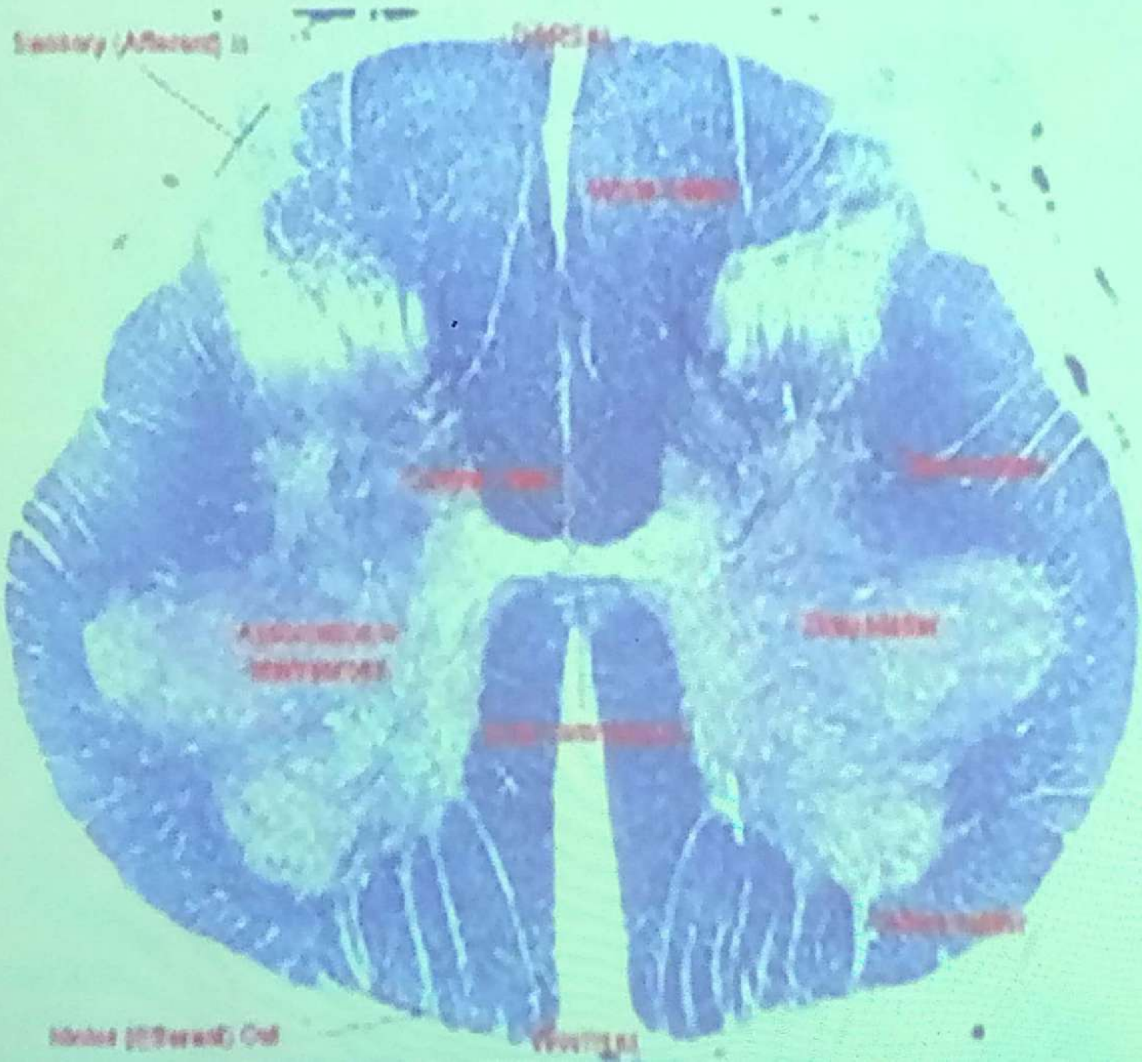
✓ **The gray matter is formed of:**

1. Nerve cell bodies or nuclei.
 They are sensory or motor.
2. Nerve fibers.
 They are dendrites and proximal naked unmyelinated axons.
3. Neuroglia.
4. Blood capillaries (non-fenestrated).

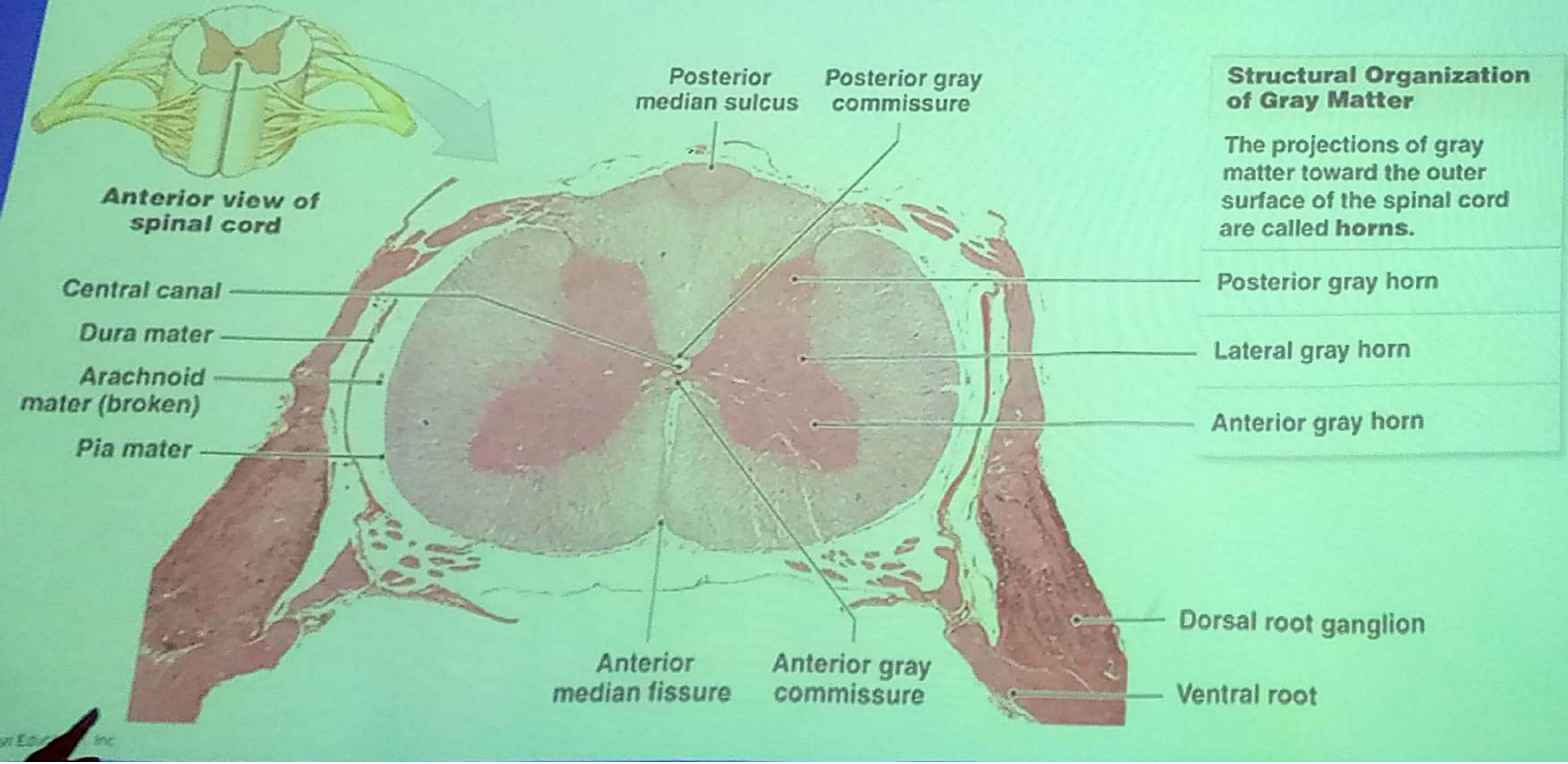
✓ **The white matter is formed of:**

1. Nerve fibers or tracts. They are myelinated.
2. Neuroglia.
3. blood capillaries (non-fenestrated).

# Structure Of The Spinal Cord



## A cross section showing most of the anatomical landmarks of the spinal cord

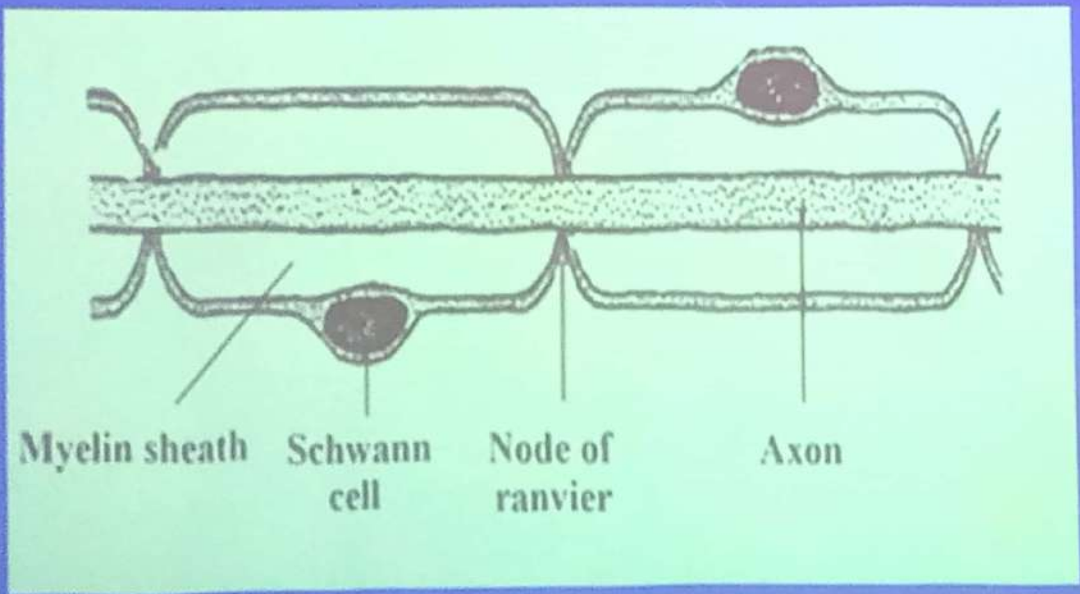


# Peripheral Nervous System (PNS)

- ✓ The PNS is formed from:
  1- Nerves
  2- Ganglia
  3- Nerve endings
- ✓ Nerves (nerve trunks) are bundles of nerve fibers surrounded by connective tissue sheaths.
- ✓ The nerve trunk is surrounded by a dense connective tissue sheath called epineurium. Within the sheath there are bundles of nerve fibers (nerve fascicles), each of which is surrounded by a connective tissue sheath called perineurium.

# Myelinated Nerve Fibers

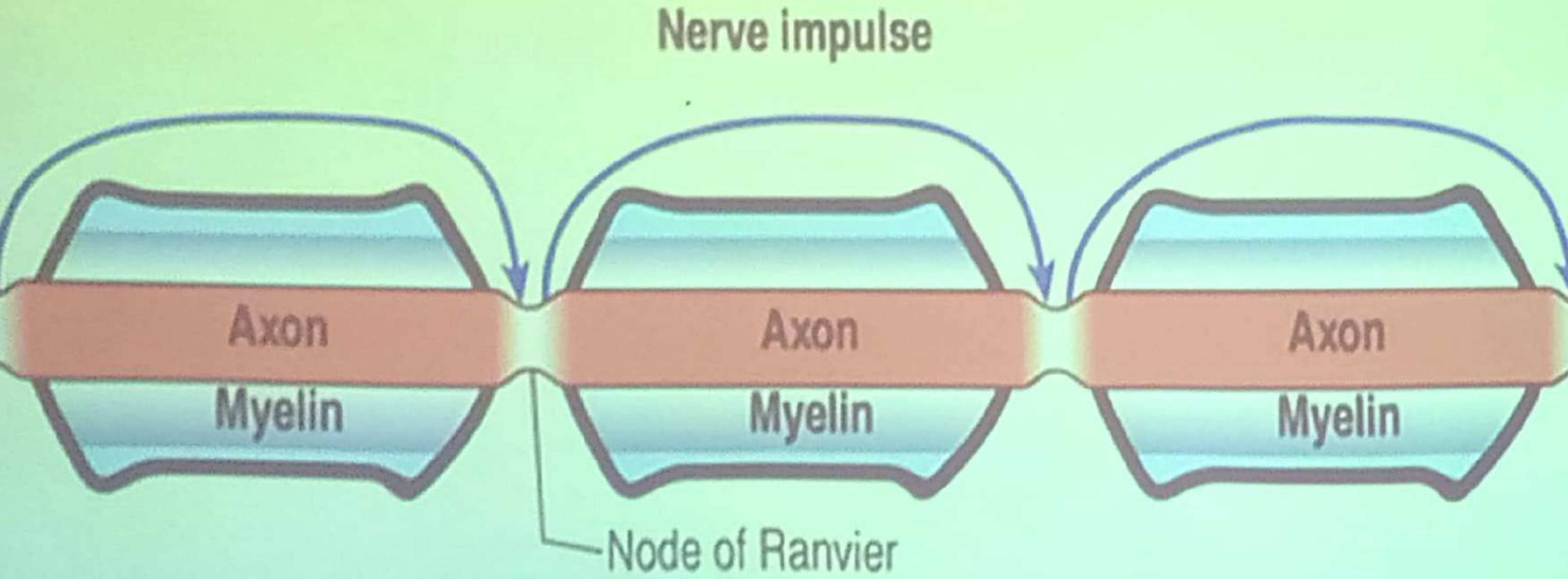
In the PNS, the myelinated nerve fiber is consisted of axon enveloped by an inner thick myelin sheath interrupted by nodes of Ranvier and an outer Schwann cell sheath (neurolemmal sheath).

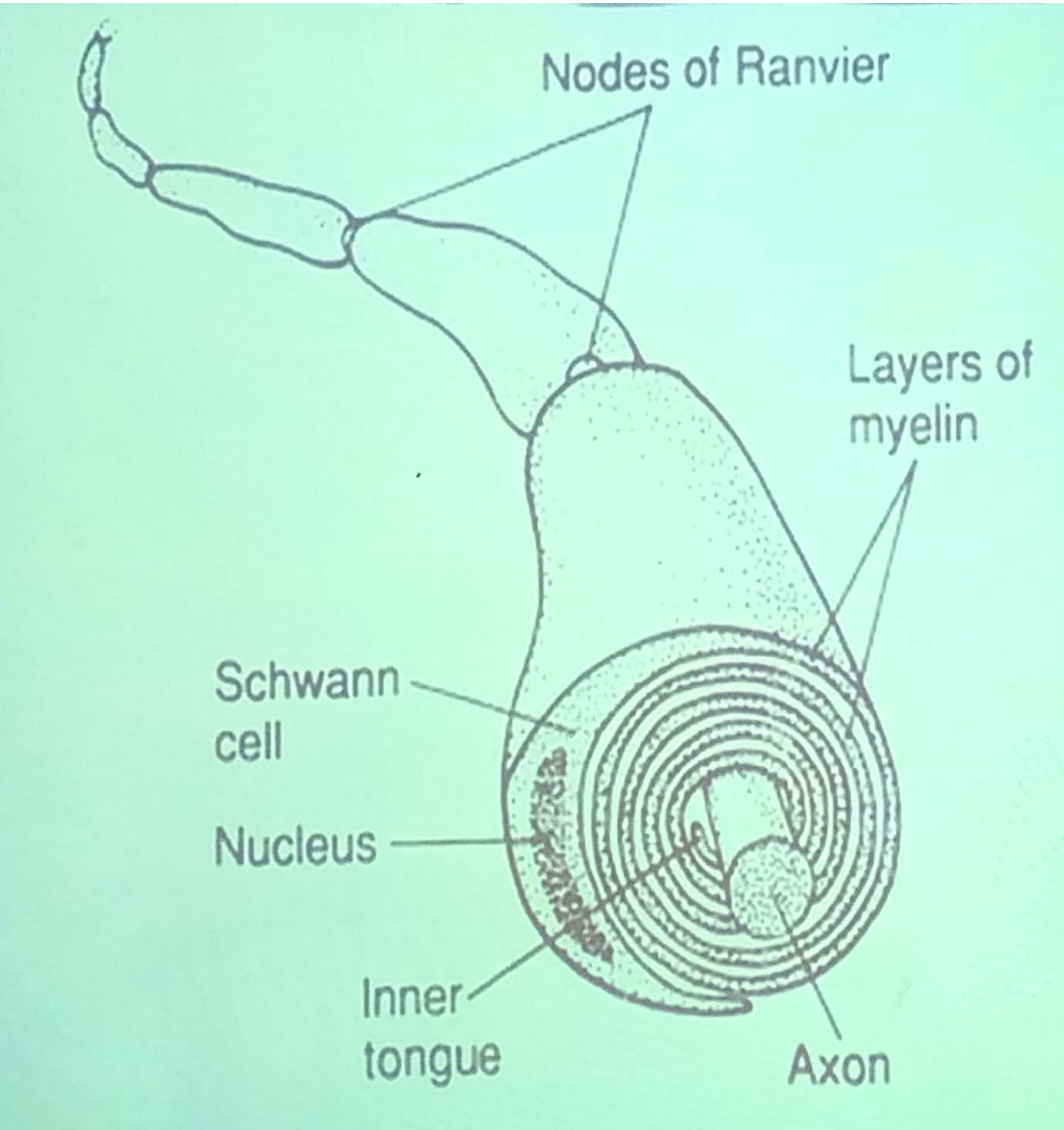


- ✓ In the CNS the myelin sheath is formed by the oligodendroglia. Cytoplasmic processes from these cells wrap the axons in a spiral manner forming the myelin sheath, while their cytoplasm stay a way from the axon and does not form a sheath as the Schwann cell does in PNS.

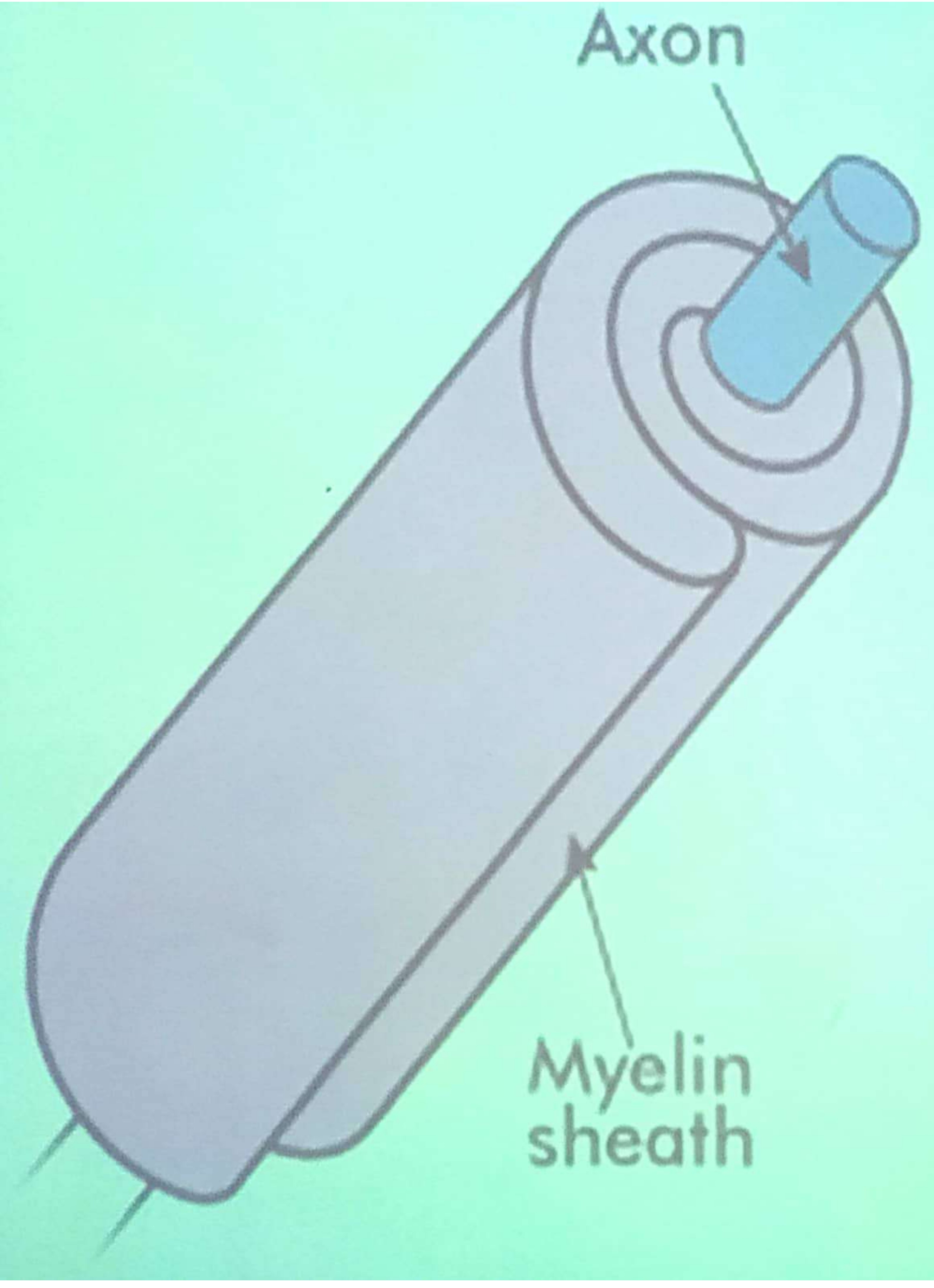
- ✓ Branches of one oligodendrocyte can envelop segments of several axons, meanwhile the Schwann cell form myelin for one segment of one axon

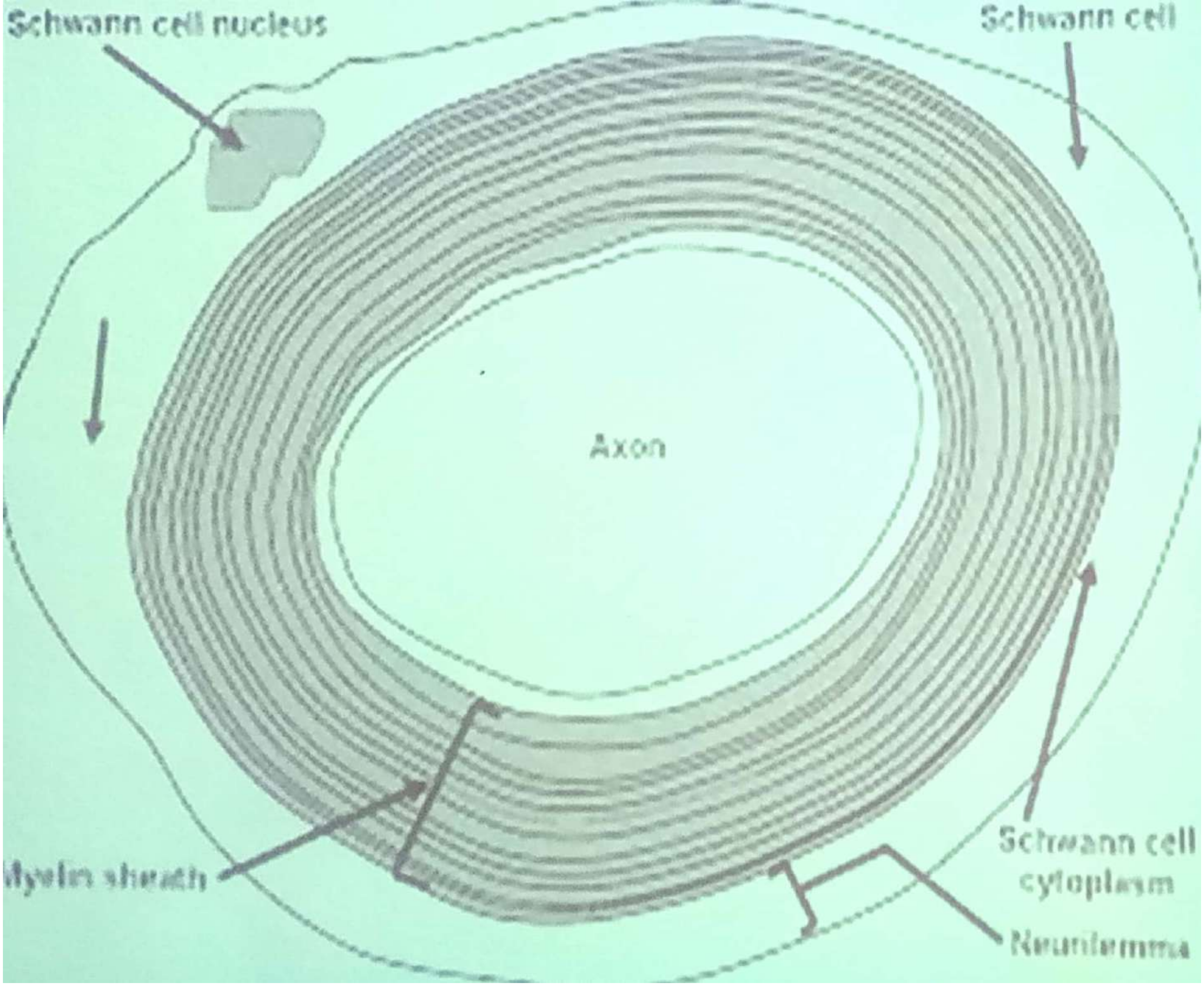
# Normal Conduction of Myelinated Nerve Fibers



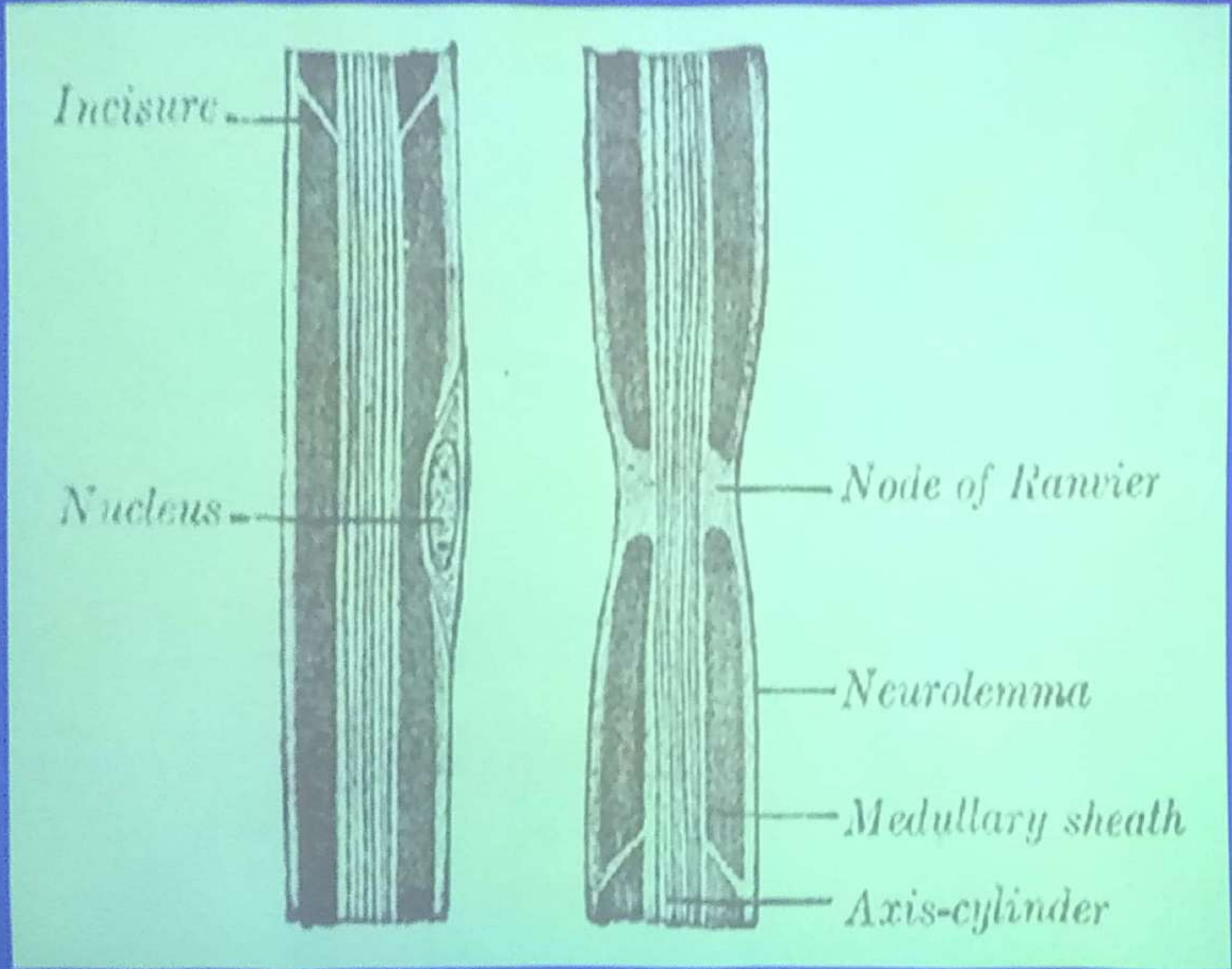


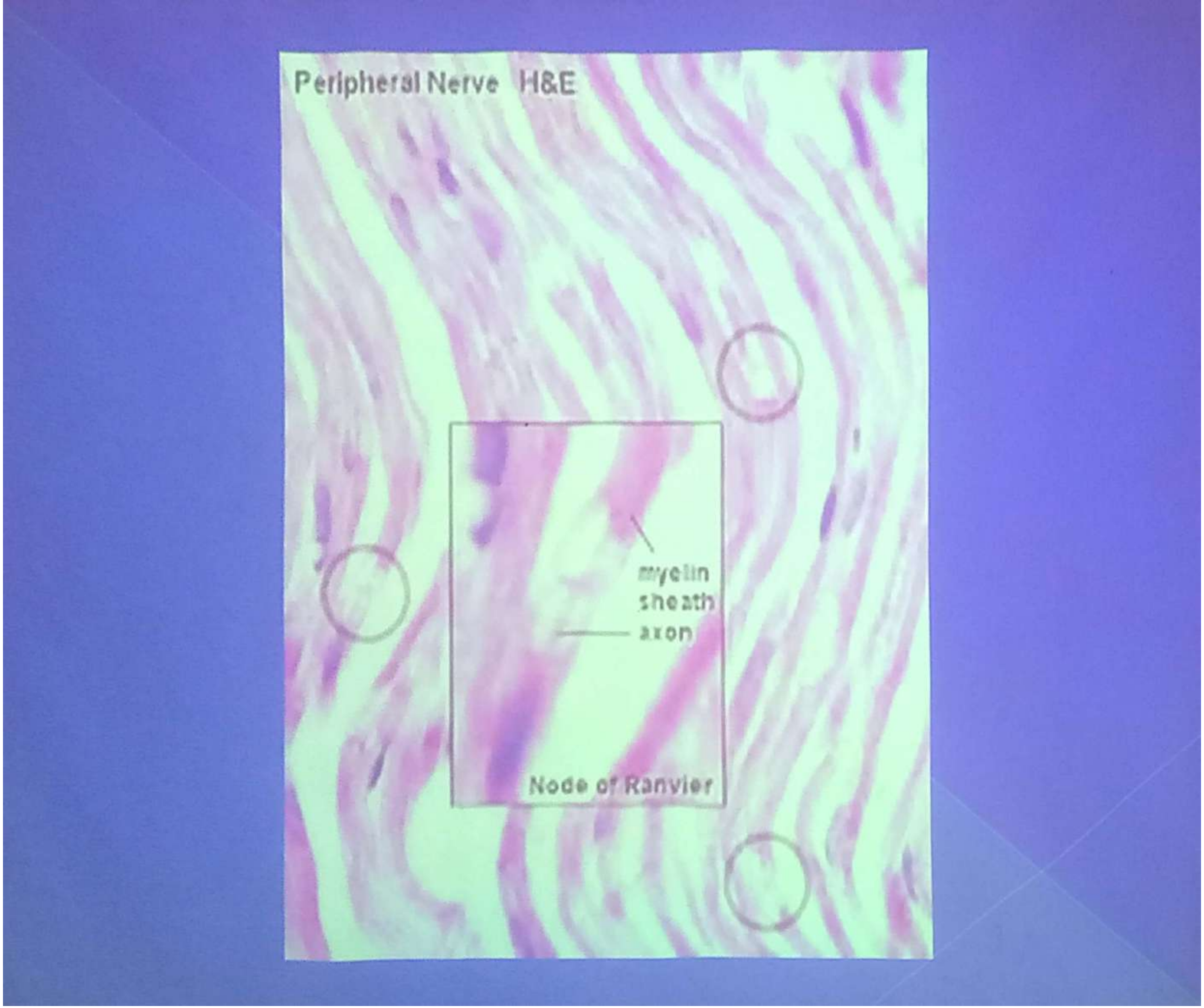
Nodes of Ranvier
Layers of myelin
Schwann cell
Nucleus
Inner tongue
Axon


Axon
Myelin sheath



Schwann cell nucleus
Schwann cell
Axon
Myelin sheath
Schwann cell
cytoplasm
Neurilemma

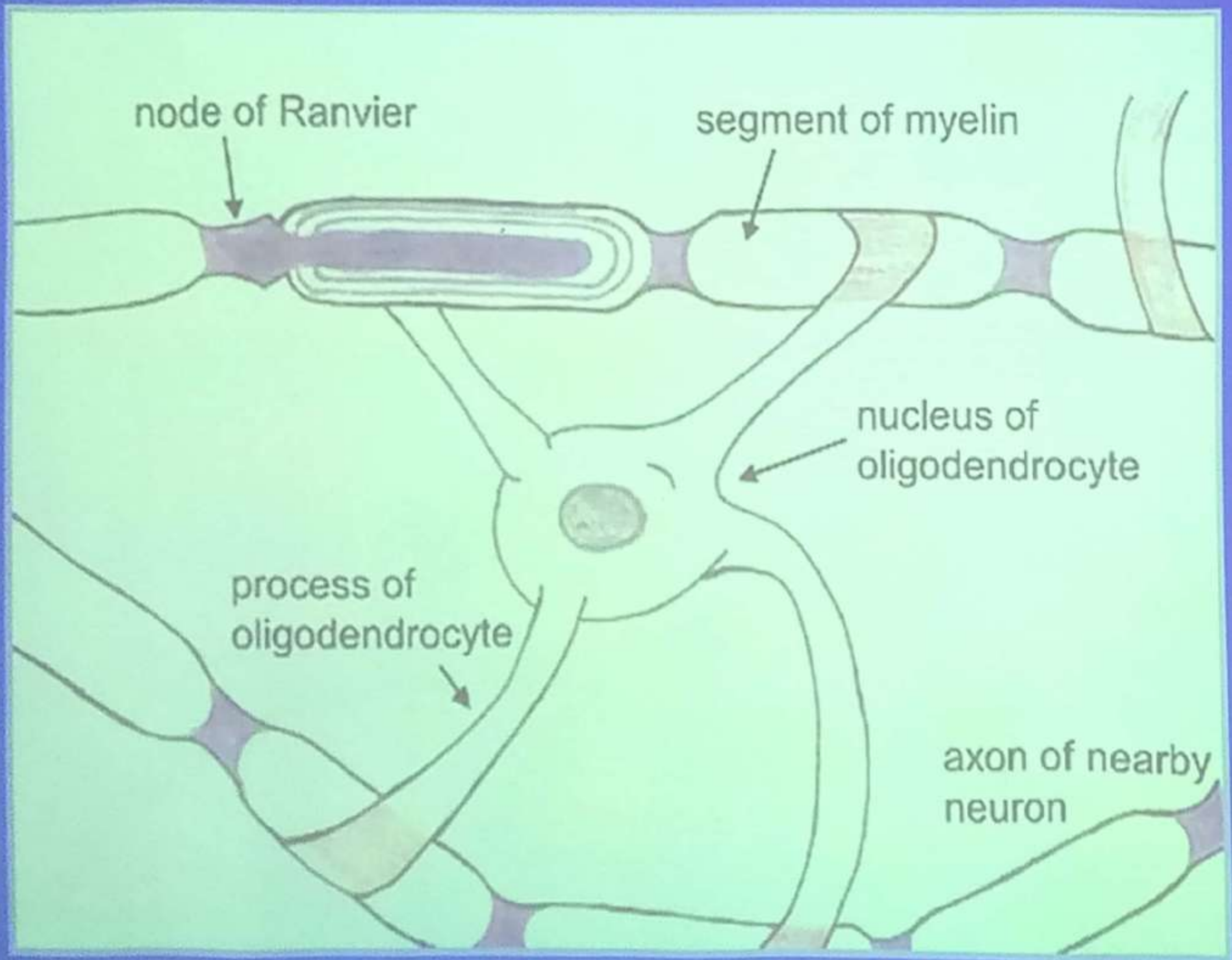

Incisure
Nucleus
Node of Ranvier
Neurolemma
Medullary sheath
Axis-cylinder


Peripheral Nerve H&E
myelin
sheath
axon
Node of Ranvier

# Unmyelinated Nerve Fibers

- ✓ In both the central and peripheral nervous systems, not all axons are sheathed in myelin.
- ✓ In PNS, all unmyelinated axons are enveloped within simple clefts of the Schwann cell.
- ✓ Each schwann cell can ensheath many unmyelinated axons.
- ✓ Unmyelinated nerve fibers do not have nodes of Ranvier, since Schwann cells are united to form a continuous sheath.

✓ The CNS is rich in unmyelinated axons, unlike those in the PNS, these axons are not sheathed.



## 2- Ganglia

- ✓ Ganglia are ovoid structures containing nerve cell bodies, neuroglia, nerve fibers, covered by connective tissue capsule and located outside the CNS.

- ✓ They serve as relay stations to transmit nerve impulses.

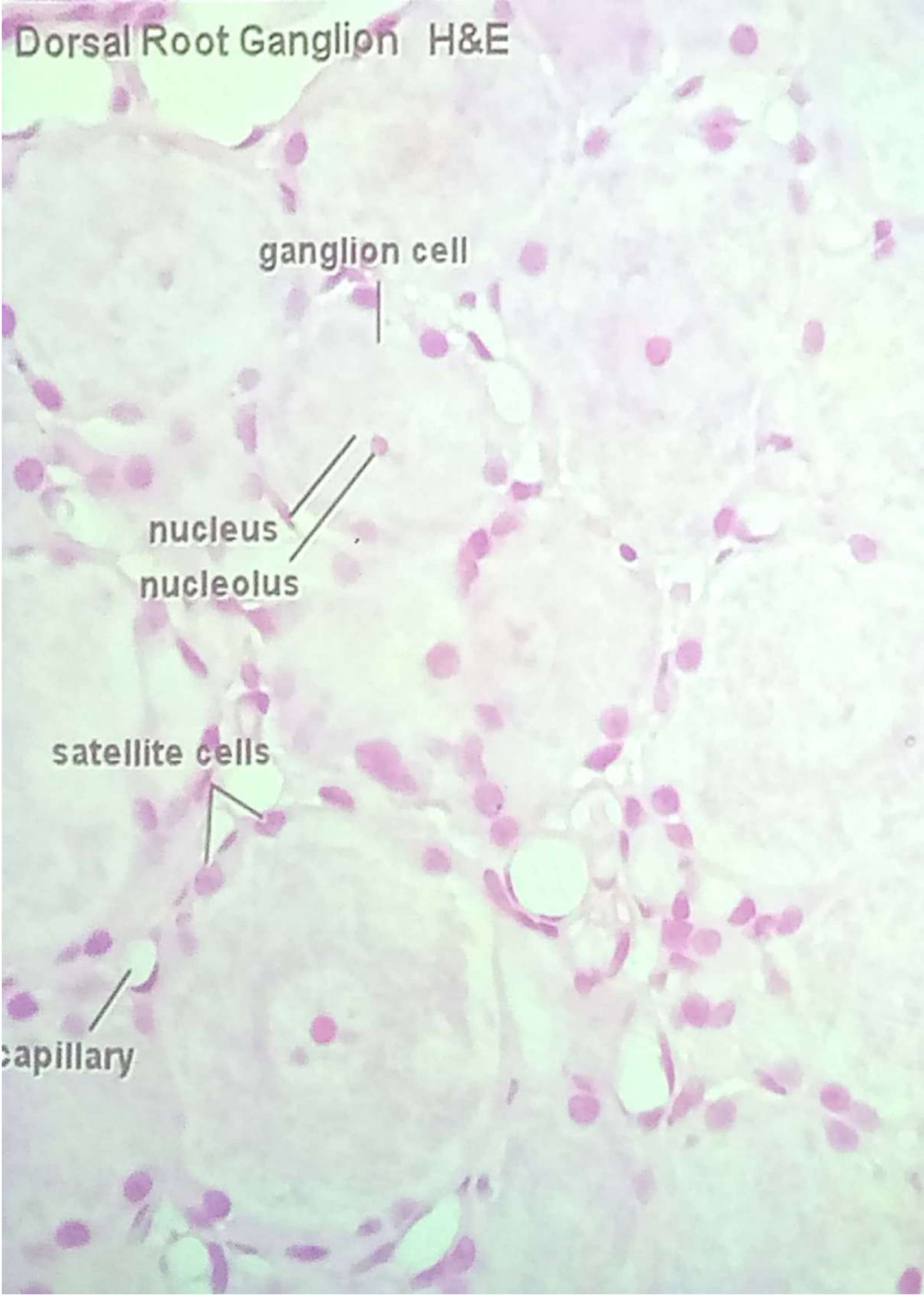
### Types of ganglia:

1- Cerebrospinal ganglia:

- a. Cerebral (cranial) ganglia.
- b. Spinal ganglia.

2- Autonomic ganglia:

- a. Sympathetic ganglia
- b. Parasympathetic ganglia


Dorsal Root Ganglion H&E
ganglion cell
nucleus
nucleolus
satellite cells
capillary


Autonomic Ganglion H&E
satellite cells
ganglion cells
lipofuscin granules

## 3- Nerve Endings

- These comprise:
  1- Receptors (sensory)
  2- Effectors (motor)
- In general all nerve terminal (sensory or motor) lose their myelin sheath and schwann cell sheath just before their terminal regions.

## I. Receptors

(Sensory nerve endings)

- Classification of receptors can be made on more than one base.